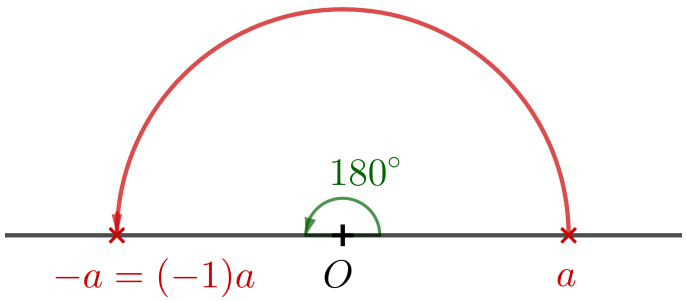
MATH1025B
Essential Mathematical Methods

Chapter 6: Complex Numbers

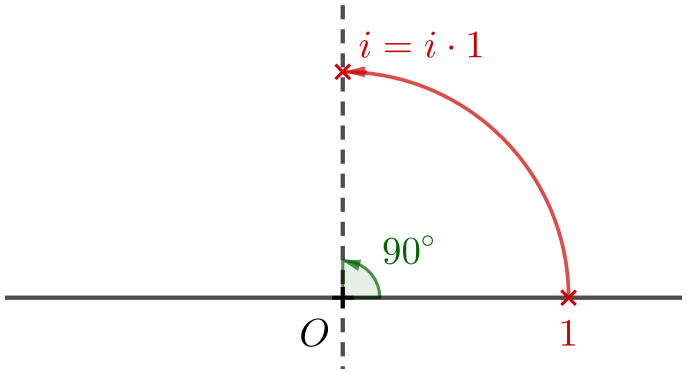
October 21, 2025

Introduction

- The quadratic equation $x^2 + 1 = 0$ has no real root.
  To "solve" it, create a new "number" i such that $i^2 = -1$.
- Geometrically, multiplying -1 to a real number a is rotating a in the anticlockwise direction by $180°$.



- $i^2 = -1$ and so $(-1)a = i \cdot i \cdot a$, which suggests that multiplying i to a is rotating a in the anticlockwise (usual convention) direction by $90°$



- In particular, $i \cdot 1 = i$. It suggests that we should jump from the real line to the plane and regard i as $(0, 1)$.

Introduction

- Each point (x, y) corresponds to a "complex number", written formally as
$$x + iy.$$
(So far, $+$ and $\cdot$ in $x + i \cdot y$ have not been defined yet.)
Then, a real number x can be regarded as a complex number $x + i \cdot 0$, lying on the x-axis.
- Furthermore, pretend we can do the following:
$$\begin{aligned}(x_1 + iy_1) + (x_2 + iy_2) &= x_1 + iy_1 + x_2 + iy_2 \\ &= x_1 + x_2 + iy_1 + iy_2 \\ &= (x_1 + x_2) + i(y_1 + y_2)\end{aligned}$$
and
$$\begin{aligned}(x_1 + iy_1) \cdot (x_2 + iy_2) &= x_1x_2 + ix_2y_1 + ix_1y_2 + iy_1iy_2 \\ &= (x_1x_2 + i^2y_1y_2) + i(x_1y_2 + x_2y_1) \\ &= (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)\end{aligned}$$

The above discussion motivates us to define a complex number as follows.

Definition 6.1

A **complex number** is defined as an ordered pair (x, y) where $x, y \in \mathbb{R}$. Addition and multiplication of complex numbers are defined as follows:

(i) (Addition)

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

(ii) (Multiplication)

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1 \cdot x_2 - y_1 \cdot y_2, x_1 \cdot y_2 + x_2 \cdot y_1)$$

A real number x corresponds to $(x, 0)$. Instead of $(0, 1)$, we write i.

We intend to write a complex number (x, y) as $x + iy$, but we need to check that it is consistent with the addition and multiplication defined above. Indeed, x, y and i correspond to $(x, 0)$, $(y, 0)$ and $(0, 1)$, so

$$\begin{aligned}(x, 0) + (0, 1) \cdot (y, 0) &= (x, 0) + (0 \cdot y - 1 \cdot 0, 0 \cdot 0 + y \cdot 1) \\ &= (x, 0) + (0, y) \\ &= (x + 0, 0 + y) \\ &= (x, y)\end{aligned}$$

From now on, a complex number (x, y) will be denoted by $x + iy$.

Introduction

- The set of all complex numbers $\{x + iy : x, y \in \mathbb{R}\}$ is denoted by $\mathbb{C}$.
- Let $z = x + iy \in \mathbb{C}$, where $x, y \in \mathbb{R}$.
  - x is said to be the real part of z, and is denoted by $\mathrm{Re}(z)$
  - y is said to be the imaginary part of z, and is denoted by $\mathrm{Im}(z)$.
- If $\mathrm{Im}(z) = 0$, z is real.
  If $\mathrm{Re}(z) = 0$, $z = iy$ which is said to be a **purely imaginary** number.
- Suppose that $w = a + ib, z = c + id \in \mathbb{C}$, where $a, b, c, d \in \mathbb{R}$. Then

$$\begin{aligned} w = z &\iff (a, b) = (c, d) \\ &\iff a = c \text{ and } b = d, \text{ i.e. } \mathrm{Re}(z) = \mathrm{Re}(w) \text{ and } \mathrm{Im}(z) = \mathrm{Im}(w) \end{aligned}$$

- $\mathbb{R}$ is regarded as a subset of $\mathbb{C}$ by the identification $\varphi : \mathbb{R} \to \mathbb{C}$ defined by $\varphi(x) = x + i \cdot 0$.
  The addition $+$ and multiplication $\cdot$ on $\mathbb{C}$ generalize those on $\mathbb{R}$ in the sense that for all $a, b \in \mathbb{R}$,

$$\varphi(a + b) = \varphi(a) + \varphi(b) \quad \text{and} \quad \varphi(a \cdot b) = \varphi(a) \cdot \varphi(b).$$

Proposition 6.1 (Properties of complex addition and multiplication)

(i) *(Commutativity of addition) For all* $w, z \in \mathbb{C}$, $w + z = z + w$.

(ii) *(Associativity of addition) For all* $u, w, z \in \mathbb{C}$, $(u + w) + z = u + (w + z)$.

(iii) *(Existence of additive identity) Let* $0 = 0 + i0 \in \mathbb{C}$. *Then, for all* $z \in \mathbb{C}$, $z + 0 = 0 + z = z$.

(iv) *(Existence of additive inverse) For all* $z = x + iy \in \mathbb{C}$, *where* $x, y \in \mathbb{R}$, $w := (-x) + i(-y) \in \mathbb{C}$ *satisfies* $z + w = w + z = 0$. w *is said to be an* ***additive inverse*** *of* z *and is denoted by* $-z$.

(v) *(Commutativity of multiplication) For all* $w, z \in \mathbb{C}$, $w \cdot z = z \cdot w$.

(vi) *(Associativity of multiplication) For all* $u, w, z \in \mathbb{C}$, $(u \cdot w) \cdot z = u \cdot (w \cdot z)$.

(vii) *(Existence of multiplicative identity) Let* $1 = 1 + i0 \in \mathbb{C}$. *Then, for all* $z \in \mathbb{C}$, $z \cdot 1 = 1 \cdot z = z$.

(viii) *(Existence of multiplicative inverse) For all nonzero* $z = x + iy \in \mathbb{C}$, *where* $x, y \in \mathbb{R}$ *with* $x \neq 0$ *or* $y \neq 0$, $w := \frac{x}{x^2+y^2} + i\frac{-y}{x^2+y^2} \in \mathbb{C}$ *satisfies* $z \cdot w = w \cdot z = 1$. w *is said to be a* ***multiplicative inverse*** *of* z *and is denoted by* z^{-1}.

(ix) *(Distributivity) For all* $u, w, z \in \mathbb{C}$, $(u + w) \cdot z = u \cdot z + w \cdot z$ *and* $z \cdot (u + w) = z \cdot u + z \cdot w$.

Remark: (i)–(ix) say that $\mathbb{C}$ is a field.

Proof of (v).
Let $z = a + ib, w = c + id \in \mathbb{C}$, where $a, b, c, d \in \mathbb{R}$.

$$\begin{aligned} z \cdot w &= (a \cdot c - b \cdot d) + i(a \cdot d + b \cdot c) && \text{(Definition of complex multiplication)} \\ &= (c \cdot a - d \cdot b) + i(d \cdot a + c \cdot b) && \text{(Commutativity of real multiplication)} \\ &= w \cdot z. \end{aligned}$$

□

Proof of (viii).
Let $z = x + iy$, $w = \frac{x}{x^2+y^2} + i\frac{-y}{x^2+y^2}$, where $x, y \in \mathbb{R}$ with $x \neq 0$ or $y \neq 0$.

$$z \cdot w = \frac{x^2 + y^2}{x^2 + y^2} + i\frac{-xy + xy}{x^2 + y^2} = 1 + i \cdot 0 = 1.$$

□

In particular, (i)–(ix) show that the usual arithmetic rules are true on $(\mathbb{C}, +, \cdot)$, for examples

$$\text{if } z \in \mathbb{C}, \text{ then } z \cdot 0 = 0 \cdot z = 0; \text{ and}$$

$$\text{if } z, w \in \mathbb{C} \text{ and } z \cdot w = 0, \text{ then } z = 0 \text{ or } w = 0.$$

Introduction

Suppose that $z = a + ib, w = c + id \in \mathbb{C}$, where $a, b, c, d \in \mathbb{R}$.
(Complex Subtraction)

$$z - w := z + (-w) = (a - c) + i(b - d)$$

(Complex Division) If $w \neq 0$,

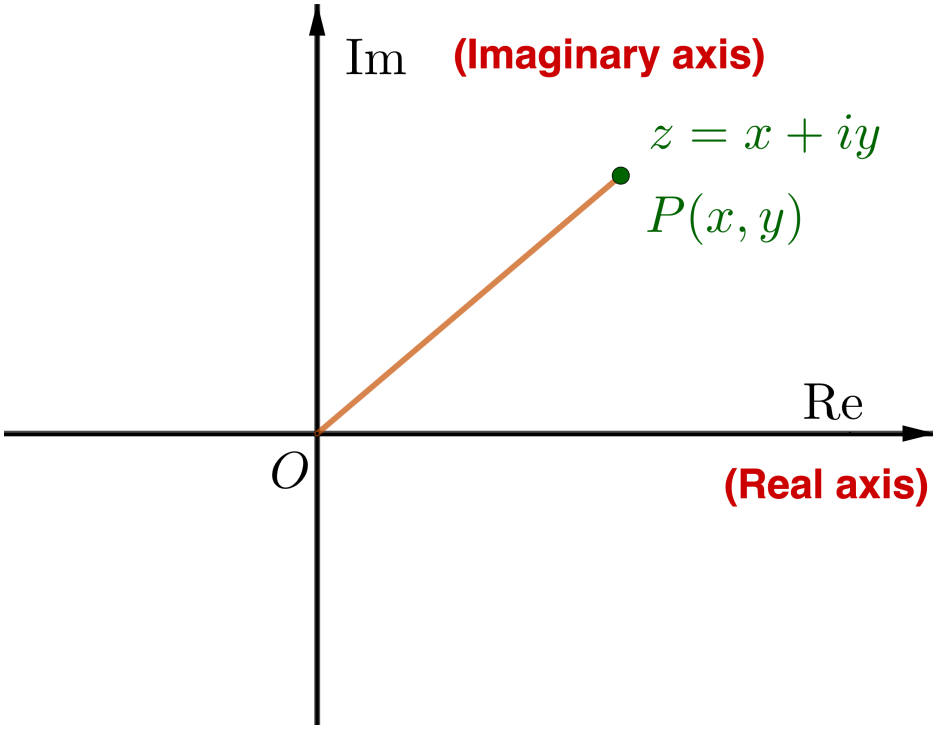
$$\frac{z}{w} := zw^{-1} = (a + ib) \cdot \left(\frac{c}{c^2 + d^2} + i \frac{-d}{c^2 + d^2} \right) = \frac{ac + bd}{c^2 + d^2} + i \frac{bc - ad}{c^2 + d^2}$$

Example 6.1

Express $\dfrac{-1 + 4i}{(2 + i) - (4 - 4i)}$ in the form $a + bi$.

$$\begin{aligned} \frac{-1 + 4i}{(2 + i) - (4 - 4i)} &= \frac{-1 + 4i}{6 - 3i} = (-1 + 4i) \cdot \left(\frac{6}{6^2 + (-3)^2} - \frac{-3}{6^2 + (-3)^2} i \right) \\ &= \frac{1}{45} (-6 - 12 + 24i - 3i) = -\frac{2}{5} + \frac{7}{15} i. \end{aligned}$$

Geometric Perspective



- A complex number $z = x + iy$ can be represented by the point or vector (x, y) on the coordinate plane.
- The plane is called the **complex plane** or **Argand's diagram**.
- The horizontal axis is called the **real axis** and the vertical axis is called the **imaginary axis**.

Complex Conjugate and Modulus

Definition 6.2

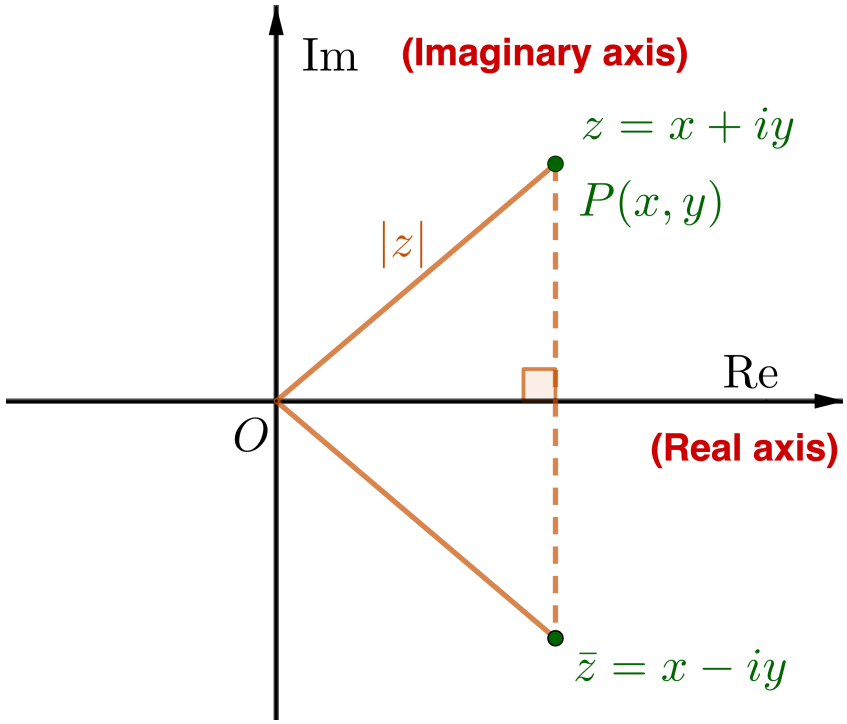
Let $z = x + iy \in \mathbb{C}$.

(i) The **complex conjugate** of z is defined to be the complex number $x - iy$. It is denoted by $\bar{z}$.

(ii) The **modulus** of z is defined to be the non-negative real number $\sqrt{x^2 + y^2}$. It is denoted by $|z|$.

Remark:

- In the complex plane, $\bar{z}$ is the reflection of z with respect to the real axis.
- $|z|$ is the length of OP.



Proposition 6.2 (Properties of complex conjugate)

(i) $\bar{\bar{z}} = z$

(ii) $z = \bar{z}$ *if and only if* z *is real*

(iii) $\mathrm{Re}(z) = \frac{1}{2}(z + \bar{z})$ *and* $\mathrm{Im}(z) = \frac{1}{2i}(z - \bar{z})$

(iv) $\overline{z_1 \pm z_2} = \overline{z_1} \pm \overline{z_2}$, $\overline{z_1 z_2} = \overline{z_1}\ \overline{z_2}$, $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z_1}}{\overline{z_2}}$.

Proof of $\overline{z_1 z_2} = \overline{z_1}\ \overline{z_2}$ in (iv).
Let $z_1 = a + ib$ and $z_2 = c + id$. Then

$$\begin{aligned}
\overline{z_1 z_2} &= \overline{(ac - bd) + i(ad + bc)} \\
&= (ac - bd) - i(ad + bc) \\
&= [ac - (-b)(-d)] + i[a(-d) + c(-b)] \\
&= (a - ib)(c - id) \\
&= \overline{z_1}\ \overline{z_2}.
\end{aligned}$$

□

Proposition 6.3 (Properties of modulus)

(i) $|z|^2 = z\bar{z}$ *(Hence, $z\bar{z}$ is real and non-negative.)*

(ii) $|z| = |-z| = |\bar{z}|$

(iii) $-|z| \leq \mathrm{Re}(z), \mathrm{Im}(z) \leq |z|$

(iv) $|z_1 z_2| = |z_1||z_2|$ *and* $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$

Proof of (iv).

$$|z_1 z_2|^2 = (z_1 z_2)\overline{(z_1 z_2)} = (z_1\overline{z_1})(z_2\overline{z_2}) = |z_1|^2|z_2|^2 = (|z_1||z_2|)^2.$$

Since modulus is always non-negative, we have $|z_1 z_2| = |z_1||z_2|$. $\square$

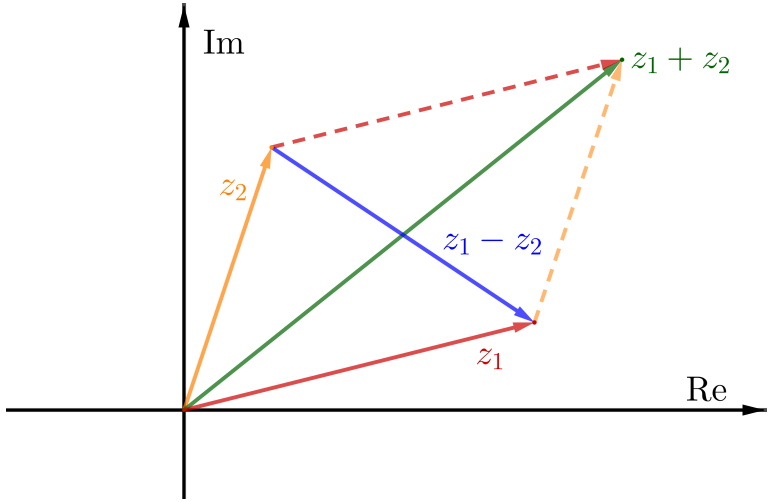
Example 6.2

Let $z_1, z_2 \in \mathbb{C}$. Show that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2$.

(Don't always write $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$!)

$$\begin{aligned}
|z_1 + z_2|^2 + |z_1 - z_2|^2 &= (z_1 + z_2)\overline{(z_1 + z_2)} + (z_1 - z_2)\overline{(z_1 - z_2)} \\
&= (z_1 + z_2)(\overline{z_1} + \overline{z_2}) + (z_1 - z_2)(\overline{z_1} - \overline{z_2}) \\
&= (z_1\overline{z_1} + z_1\overline{z_2} + z_2\overline{z_1} + z_2\overline{z_2}) + (z_1\overline{z_1} - z_1\overline{z_2} - z_2\overline{z_1} + z_2\overline{z_2}) \\
&= 2z_1\overline{z_1} + 2z_2\overline{z_2} \\
&= 2|z_1|^2 + 2|z_2|^2.
\end{aligned}$$

This result is called the parallelogram law.



Exercise 6.1

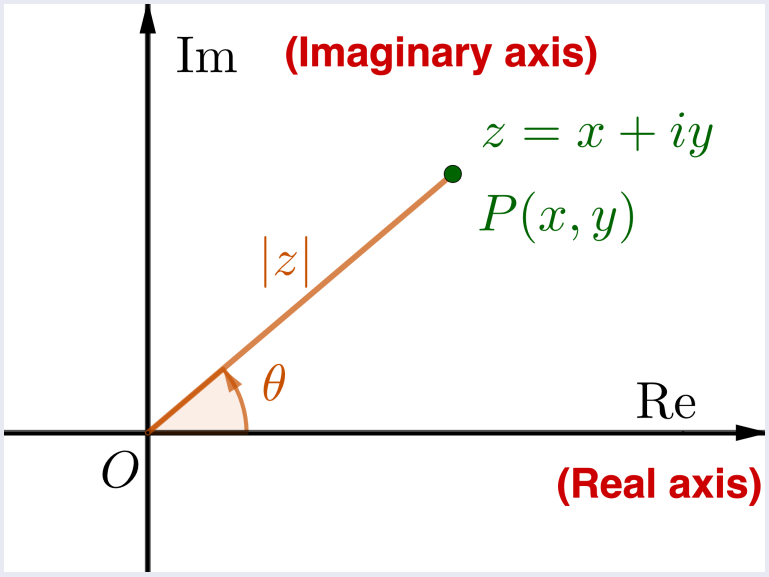
Let $z_1, z_2 \in \mathbb{C}$. Show that $|1 - z_1\overline{z_2}|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$.

Exercise 6.2

Let $z \in \mathbb{C}$ such that $z \neq 0$. Show that $\left|\bar{z} + \frac{1}{z}\right| \geq 2$

Definition 6.3

Let $z = x + iy$ be a non-zero complex number.



- θ is called an **argument** of z, denoted by $\arg(z)$, if θ is an angle at the origin measured anticlockwise from the positive real axis to the point (x, y). Note that $\arg(z)$ is only determined up to $2k\pi$, $k \in \mathbb{Z}$, additively:
$$\arg(z) = \theta + 2k\pi, \quad k \in \mathbb{Z}.$$
- An argument θ of z is said to be the **principal argument** of z if $-\pi < \theta \leq \pi$. It is denoted by $\mathrm{Arg}(z)$.

Polar Form

A non-zero complex number z can be expressed as

$$z = r(\cos\theta + i\sin\theta) \quad (\text{or simply } r\operatorname{cis}\theta),$$

where

$$r = |z| \quad \text{and} \quad \theta \text{ is an argument of } z.$$

It is called the **polar form** of z.

Remark: If $z = r(\cos\theta + i\sin\theta)$ for some $r > 0$ and $\theta \in \mathbb{R}$, then

- $|z| = r$ and θ is an argument of z.
- $\bar{z} = r(\cos\theta - i\sin\theta) = r\,[\cos(-\theta) + i\sin(-\theta)]$ (Polar form)
- $\dfrac{1}{z} = \dfrac{1}{z\bar{z}}\bar{z} = \dfrac{1}{|z|^2}\bar{z} = \dfrac{1}{r}[\cos(-\theta) + i\sin(-\theta)]$ (Polar form)

  In particular, $|z| = 1 \iff \dfrac{1}{z} = \bar{z}$.

Example 6.3

Let $z = -2 + 2\sqrt{3}i$. Find the modulus, argument and principal argument of z. Express z in polar form.

Modulus: $|z| = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{4+12} = \sqrt{16} = 4$.

Solving $\begin{cases} 4\cos\theta = -2 \\ 4\sin\theta = 2\sqrt{3} \end{cases}$, we have $\tan\theta = -\sqrt{3}$ and so

$$\theta = -\frac{\pi}{3} \text{ (rejected)} \quad \text{or} \quad \theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}.$$

Principal argument: $\operatorname{Arg}(z) = \dfrac{2\pi}{3}$.

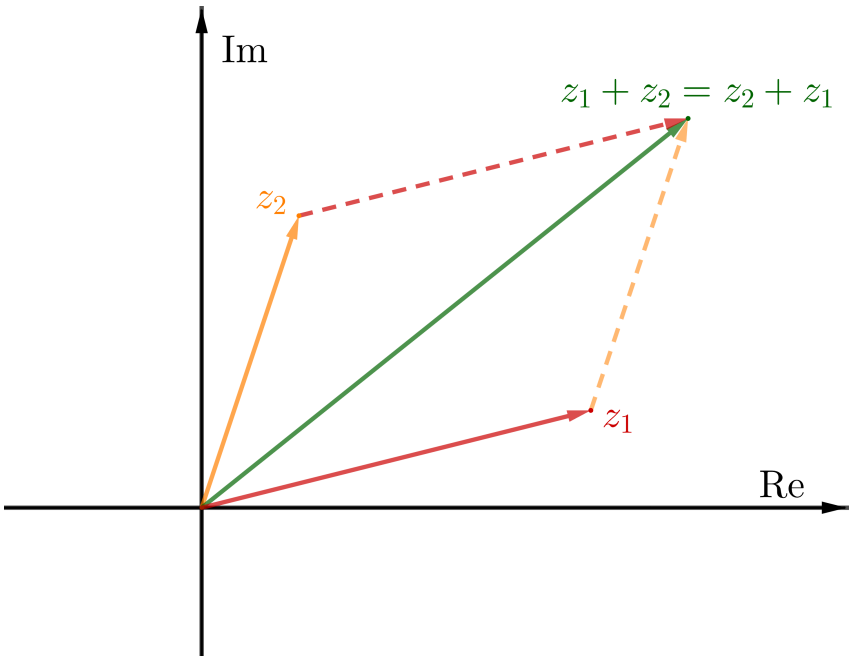
Argument: $\arg(z) = \dfrac{2\pi}{3} + 2k\pi$, $k \in \mathbb{Z}$.

Polar form: $z = 4\left(\cos\dfrac{2\pi}{3}) + i\sin(\dfrac{2\pi}{3})\right)$.

Visualization of Complex Addition and Multiplication

Let $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2 \in \mathbb{C}$, where $x_1, x_2, y_1, y_2 \in \mathbb{R}$.
Complex addition can be visualized similar to vector addition in $\mathbb{R}^2$.

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$



How about complex multiplication?

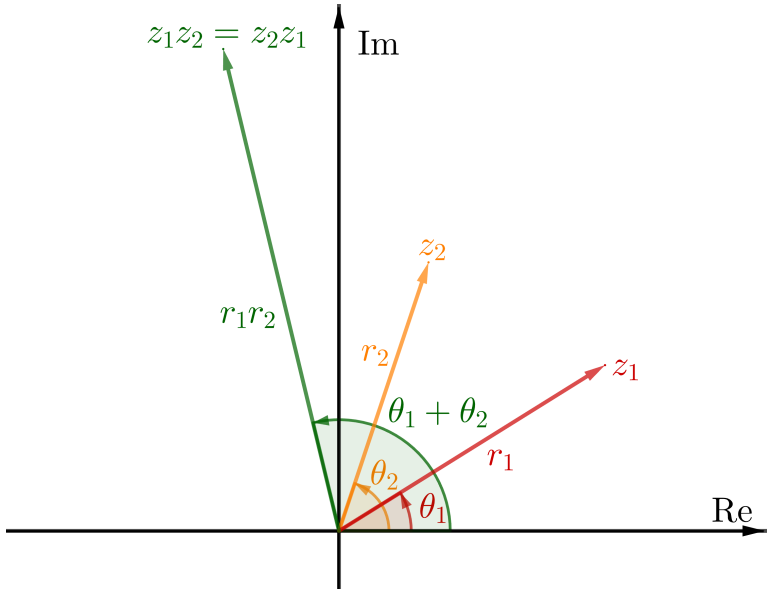
$$z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$$

which may not give us any idea.

Visualization of Complex Addition and Multiplication

Let's try polar form.
Assume $z_1, z_2 \neq 0$. Write $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$, where $r_1, r_2 > 0$, $\theta_1, \theta_2 \in \mathbb{R}$. Then

$$\begin{aligned} z_1 z_2 &= r_1 r_2 \left[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i\,(\cos\theta_1\sin\theta_2 + \cos\theta_2\sin\theta_1)\right] \\ &= r_1 r_2 \left[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\right]. \end{aligned}$$



We have

$$\begin{cases} |z_1 z_2| = |z_1||z_2| \\ \arg(z_1 z_2) = \arg(z_1) + \arg(z_2). \end{cases}$$

(Is $\mathrm{Arg}(z_1 z_2) = \mathrm{Arg}(z_1) + \mathrm{Arg}(z_2)$ true? Why?)

Visualization of Complex Addition and Multiplication

In particular, we have proved the following.

Proposition 6.4

Let $z = |z|(\cos\theta + i\sin\theta)$, $w = |w|(\cos\varphi + i\sin\varphi) \in \mathbb{C}$. *Then*

$$zw = |z||w|(\cos(\theta+\varphi) + i\sin(\theta+\varphi)).$$

Complex multiplication as applying a rotation and a scaling

$$zw = \underbrace{|z|}_{\text{scaling}} \underbrace{(\cos\theta + i\sin\theta)}_{\text{rotation}} w$$

Similarly,

$$\begin{aligned}\frac{z_1}{z_2} = z_1 z_2^{-1} &= r_1(\cos\theta_1 + i\sin\theta_1)\cdot\frac{1}{r_2}[\cos(-\theta_2) + i\sin(-\theta_2)] \\ &= \frac{r_1}{r_2}\left[\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2)\right].\end{aligned}$$

We have

$$\begin{cases}\left|\dfrac{z_1}{z_2}\right| = \dfrac{|z_1|}{|z_2|} \\ \arg(\dfrac{z_1}{z_2}) = \arg(z_1) - \arg(z_2).\end{cases}$$

Definition 6.4

Let $z \in \mathbb{C}$. We define $z^{n+1} := (z^n) \cdot z$ for $n \in \mathbb{Z}^+$. Further, if $z \neq 0$, we define $z^0 := 1$ and $z^{-n} := (z^{-1})^n$ for $n \in \mathbb{Z}^+$.

With this definition, we can show that the usual Laws of Indices hold for integral powers of complex numbers. Furthermore, the following holds.

Theorem 6.5 (De Moivre's Theorem)

For any $n \in \mathbb{Z}$, $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

Proof. *Case 1:* n is a positive integer. Prove by mathematical induction and Proposition 6.4.
Case 2: $n = 0$. Trivial.
Case 3: n is a negative integer. Let $m = -n$. Then m is a positive integer.

$$\begin{aligned}
(\cos\theta + i\sin\theta)^n &= \left[(\cos\theta + i\sin\theta)^{-1}\right]^m = \left[(\cos\theta + i\sin\theta)^m\right]^{-1} \\
&= (\cos m\theta + i\sin m\theta)^{-1} && \text{(by Case 1)} \\
&= \cos m\theta - i\sin m\theta && (|z| = 1 \implies z^{-1} = \bar{z}) \\
&= \cos(-m\theta) + i\sin(-m\theta) \\
&= \cos n\theta + i\sin n\theta.
\end{aligned}$$

□

Example 6.4

By De Moivre's Theorem,

$$\begin{aligned}\cos 3\theta + i\sin 3\theta &= (\cos\theta + i\sin\theta)^3 \\ &= \cos^3\theta + 3\cos^2\theta(i\sin\theta) + 3\cos\theta(i\sin\theta)^2 + (i\sin\theta)^3 \\ &= (\cos^3\theta - 3\cos\theta\sin^2\theta) + i(3\cos^2\theta\sin\theta - \sin^3\theta).\end{aligned}$$

Comparing the real and imaginary parts, we have

$$\begin{aligned}\cos 3\theta &= \cos^3\theta - 3\cos\theta\sin^2\theta = \cos^3\theta - 3\cos\theta(1-\cos^2\theta) \\ &= 4\cos^3\theta - 3\cos\theta,\end{aligned}$$

$$\begin{aligned}\sin 3\theta &= 3\cos^2\theta\sin\theta - \sin^3\theta = 3(1-\sin^2\theta)\sin\theta - \sin^3\theta \\ &= 3\sin\theta - 4\sin^3\theta.\end{aligned}$$

Exercise 6.3

Express $\cos 5\theta$ and $\sin 5\theta$ in terms of $\cos\theta$ and $\sin\theta$.

Example 6.5

Show that $8\sin^4\theta = \cos 4\theta - 4\cos 2\theta + 3$.

Let $z = \cos\theta + i\sin\theta$. Then $\begin{cases} \cos\theta = \mathrm{Re}(z) = \frac{1}{2}(z+\bar{z}) \\ \sin\theta = \mathrm{Im}(z) = \frac{1}{2i}(z-\bar{z}). \end{cases}$

By De Moivre's Theorem, $z^n = (\cos n\theta + i\sin n\theta)$ for $n \in \mathbb{Z}$. Then

$$\begin{cases} \cos n\theta = \mathrm{Re}(z^n) = \frac{1}{2}(z^n + \bar{z}^n) \\ \sin n\theta = \mathrm{Im}(z^n) = \frac{1}{2i}(z^n - \bar{z}^n). \end{cases}$$

Hence,

$$\begin{aligned} 8\sin^4\theta &= 8\left[\frac{1}{2i}(z-\bar{z})\right]^4 \\ &= \frac{1}{2}\left(z^4 - 4z^3\bar{z} + 6z^2\bar{z}^2 - 4z\bar{z}^3 + \bar{z}^4\right) \\ &= \frac{1}{2}(z^4 + \bar{z}^4) - 2(z^2 + \bar{z}^2) + 3 \qquad (\because z\bar{z} = |z|^2 = 1) \\ &= \cos 4\theta - 4\cos 2\theta + 3. \end{aligned}$$

Exercise 6.4

Show that $\cos^3\theta\sin^4\theta = \dfrac{1}{64}\left(\cos 7\theta - \cos 5\theta - 3\cos 3\theta + 3\cos\theta\right)$.

Triangle Inequality

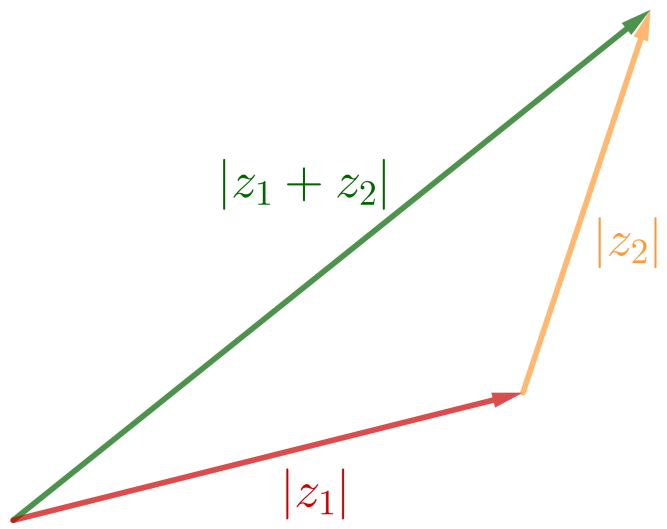
Theorem 6.6 (Triangle Inequality)

Let $z_1, z_2 \in \mathbb{C}$. *Then*

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

Furthermore, the equality holds if and only if

$$z_1 = kz_2 \quad \textit{for some } k \geq 0 \quad \textit{or} \quad z_2 = kz_1 \quad \textit{for some } k \geq 0 \ .$$



Triangle Inequality

Proof. The statement is trivially true if $z_1 = 0$ or $z_2 = 0$. Assume that $z_1, z_2 \neq 0$.

$$\begin{aligned}
|z_1 + z_2|^2 &= (z_1 + z_2)(\overline{z_1} + \overline{z_2}) \\
&= z_1\overline{z_1} + z_1\overline{z_2} + z_2\overline{z_1} + z_2\overline{z_2} \\
&= |z_1|^2 + 2\,\mathrm{Re}(z_1\overline{z_2}) + |z_2|^2 \\
&\leq |z_1|^2 + 2|z_1\overline{z_2}| + |z_2|^2 \\
&\leq |z_1|^2 + 2|z_1||z_2| + |z_2|^2 \\
&= (|z_1| + |z_2|)^2.
\end{aligned}$$

So $|z_1 + z_2| \leq |z_1| + |z_2|$.

$$\begin{aligned}
\text{Furthermore, the equality holds} \quad &\iff \quad z_1\overline{z_2} = k \quad \text{for some } k \geq 0 \\
&\qquad\quad z_1 = \frac{k}{\overline{z_2}} = \frac{k}{|z_2|^2} z_2 \\
&\iff \quad z_1 = k' z_2 \quad \text{for some } k' \geq 0.
\end{aligned}$$

Triangle Inequality

Exercise 6.5

Let $z_1, z_2, \ldots, z_n \in \mathbb{C}$. By mathematical induction, show that

$$\left|\sum_{i=1}^{n} z_i\right| \leq \sum_{i=1}^{n} |z_i|.$$

Exercise 6.6

Let $z_1, z_2, \ldots, z_n \in \mathbb{C}$ and $\lambda_1, \lambda_2, \ldots, \lambda_n > 0$ such that $\lambda_1^{-1} + \lambda_2^{-1} + \cdots + \lambda_n^{-1} = 1$. Show that

$$\left|\sum_{i=1}^{n} z_i\right|^2 \leq \sum_{i=1}^{n} \lambda_i |z_i|^2.$$

Euler's Formula and Complex Numbers

Recall the series expansions:

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Motivation:
Let $z = x + iy \in \mathbb{C}$, where $x, y \in \mathbb{R}$. How do we define e^z?
Assume that the laws of indices still hold:

$$e^z = e^{x+iy} = \underbrace{e^x}_{\text{real}} \cdot \underbrace{e^{iy}}_{\text{What is it?}}$$

Euler's Formula and Complex Numbers

Pretending the series expansions are true for complex number $i\theta$, we have

$$\begin{aligned} e^{i\theta} &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \\ &= 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} - \cdots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right) \\ &= \cos\theta + i\sin\theta. \end{aligned}$$

It suggests us to define

$$e^{i\theta} = \cos\theta + i\sin\theta \qquad \text{for } \theta \in \mathbb{R}.$$

Then, we can write a complex number z as

$$z = re^{i\theta},$$

where $r = |z|$ and θ is an argument of z (when $z \neq 0$).
This is called the **exponential form** of z.

Euler's Formula for Complex Numbers

By expressing complex numbers in the exponential form, the complex multiplication and De Moibre's Theorem can be written as follows:

$$\text{Multiplication:} \quad z_1 z_2 = r_1 e^{i\theta} \cdot r_2 e^{i\theta} = r_1 r_2 e^{i(\theta_1+\theta_2)}$$

$$\text{De Moivre's Theorem:} \quad (e^{i\theta})^n = e^{in\theta}, \quad n \in \mathbb{Z}.$$

Also,

$$\cos\theta = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right), \qquad \sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right).$$

Putting $\theta = \pi$ yields

$$e^{i\pi} = \cos\pi + i\sin\pi = -1 + i(0) = -1.$$

Euler's Identity

$$e^{i\pi} + 1 = 0$$

It connects five fundamental numbers in mathematics.

Example 6.6

$$\begin{aligned}
\int \sin^2\theta\cos^2\theta\,d\theta &= \int \left[\frac{1}{2i}(e^{i\theta}-e^{-i\theta})\right]^2 \left[\frac{1}{2}(e^{i\theta}+e^{-i\theta})\right]^2 d\theta \\
&= -\frac{1}{16}\int (e^{2i\theta}-2+e^{-2i\theta})(e^{2i\theta}+2+e^{-2i\theta})\,d\theta \\
&= -\frac{1}{16}\int (e^{4i\theta}+e^{-4i\theta}-2)\,d\theta \\
&= -\frac{1}{16}\left[\frac{1}{4i}(e^{4i\theta}-e^{-4i\theta})-2\theta\right]+C \qquad (*) \\
&= -\frac{1}{32}\sin 4\theta+\frac{\theta}{8}+C.
\end{aligned}$$

In $(*)$, we assume that for $\alpha \in \mathbb{C}\backslash\{0\}$,

$$\int e^{\alpha x}\,dx = \frac{1}{\alpha}e^{\alpha x}+C.$$

Example 6.7

Let θ be a real number such that $\theta \neq 2k\pi$, $k \in \mathbb{Z}$. Show that, for $n \in \mathbb{Z}^+$,

$$\sum_{k=1}^{n} \cos k\theta = \frac{\sin \frac{n\theta}{2} \cos \frac{(n+1)\theta}{2}}{\sin \frac{\theta}{2}}, \qquad \sum_{k=1}^{n} \sin k\theta = \frac{\sin \frac{n\theta}{2} \sin \frac{(n+1)\theta}{2}}{\sin \frac{\theta}{2}}.$$

If $z \neq 1$, then

$$\begin{aligned}(z-1)(z+z^2+\cdots+z^n) &= z^{n+1} - z \\ z+z^2+\cdots+z^n &= z \cdot \frac{z^n - 1}{z-1}.\end{aligned}$$

Now, put $z = e^{i\theta} \neq 1$,

$$\begin{aligned}\text{LHS} &= e^{i\theta} + e^{i2\theta} + \cdots + e^{in\theta} \\ &= \sum_{k=1}^{n} \cos k\theta + i \sum_{k=1}^{n} \sin k\theta,\end{aligned}$$

Euler's Formula for Complex Numbers

$$
\begin{aligned}
\text{LHS} &= \sum_{k=1}^{n} \cos k\theta + i \sum_{k=1}^{n} \sin k\theta. \\
\text{RHS} &= e^{i\theta} \cdot \frac{e^{in\theta} - 1}{e^{i\theta} - 1} \\
&= e^{i\theta} \cdot \frac{e^{i\frac{n\theta}{2}} - e^{-i\frac{n\theta}{2}}}{e^{i\frac{\theta}{2}} - e^{-i\frac{\theta}{2}}} \cdot \frac{e^{i\frac{n\theta}{2}}}{e^{i\frac{\theta}{2}}} \\
&= \frac{2i \sin \frac{n\theta}{2}}{2i \sin \frac{\theta}{2}} \cdot e^{i\frac{(n+1)\theta}{2}} \\
&= \frac{\sin \frac{n\theta}{2}}{\sin \frac{\theta}{2}} \left(\cos \frac{(n+1)\theta}{2} + i \sin \frac{(n+1)\theta}{2} \right).
\end{aligned}
$$

By comparing the real and imaginary parts, we have

$$
\sum_{k=1}^{n} \cos k\theta = \frac{\sin \frac{n\theta}{2} \cos \frac{(n+1)\theta}{2}}{\sin \frac{\theta}{2}}, \qquad \sum_{k=1}^{n} \sin k\theta = \frac{\sin \frac{n\theta}{2} \sin \frac{(n+1)\theta}{2}}{\sin \frac{\theta}{2}}.
$$

Fundamental Theorem of Algebra

Let $p(x) \in \mathbb{R}[x]$ (the set of all polynomials with coefficients in $\mathbb{R}$) be a **nonconstant** polynomial, i.e.

$$p(x) = a_n x^n + a_{n_1} x^{n-1} + \cdots + a_1 x + a_0$$

for some $n \in \mathbb{Z}^+$ and some $a_n, a_{n-1}, \ldots, a_1, a_0 \in \mathbb{R}$ with $a_n \neq 0$.
Here a_n is called the **leading coefficient** of $p(x)$ and n is called the **degree** of $p(x)$, denoted by $\deg(p(x))$.

Question

Can we always find $\alpha \in \mathbb{R}$ such that $p(\alpha) = 0$?

Answer: Negative! Consider $p(x) = x^2 + 1$. $p(x) = 0$ has no real roots!.

How about $\mathbb{C}$?

Theorem 6.7 (Fundamental Theorem of Algebra)

Let $p \in \mathbb{C}[x]$ be a nonconstant polynomial (with coefficients in $\mathbb{C}$). Then $p(x) = 0$ has at least one root in $\mathbb{C}$.

Fundamental Theorem of Algebra

Direct Consequence:
Suppose that $p(\alpha) = 0$. Then, $p(x) = (x - \alpha)q(x)$ for some $q(x) \in \mathbb{C}[x]$ where

$$\deg(q(x)) = \deg(p(x)) - 1.$$

Apply the fundamental theorem of algebra repeatedly, $p(x)$ can be factorized as

$$p(x) = a_n(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n),$$

where $\alpha_i \in \mathbb{C}$ (α_i's are not necessarily distinct).
Therefore, $p(x)$ has exactly n complex roots, counting with multiplicities.

Example 6.8

$p(x) = (1 + i)x^3 + 2x^2 + (3 + 5i)x + (6 - 2i)$ can be factorized as

$$p(x) = (1 + i)(x - i)(x + 1 - 2i)(x + 2i).$$

Fundamental Theorem of Algebra

Coming back to polynomials with real coefficients:
Consider $x^2 + 2x + 5 = 0$. Then

$$\Delta = 2^2 - 4(1)(5) = -16 < 0.$$

So it has no real root. By the quadratic formula,

$$x = \frac{-2 \pm \sqrt{-16}}{2} = -1 \pm 4i.$$

Observation:

$$\text{Both } \alpha = -1 + 4i \text{ and } \overline{\alpha} = -1 - 4i \text{ are roots.}$$

In fact, it is true in general.

Proposition 6.8

Let $p(x) \in \mathbb{R}[x]$, *i.e.*

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \qquad \textit{where } a_i \in \mathbb{R} \textit{ for } i = 0, 1, \ldots, n.$$

If $p(\alpha) = 0$, *then* $p(\overline{\alpha}) = 0$. *(Trivial if* $\alpha \in \mathbb{R}$*.)*

Proof.

$$a_n \alpha^n + a_{n-1} \alpha^{n-1} + \cdots + a_1 \alpha + a_0 = p(\alpha) = 0.$$

Taking complex conjugate on both sides,

$$\overline{a_n \alpha^n + a_{n-1} \alpha^{n-1} + \cdots + a_1 \alpha + a_0} = \overline{0}$$
$$a_n (\overline{\alpha})^n + a_{n-1} (\overline{\alpha})^{n-1} + \cdots + a_1 (\overline{\alpha}) + a_0 = 0,$$

since

$$\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}, \qquad \overline{z_1 z_2} = \overline{z_1}\,\overline{z_2} \qquad \text{and} \qquad \overline{a_i} = a_i.$$

Therefore $p(\overline{\alpha}) = 0$.

Proposition 6.8

Let $p(x) \in \mathbb{R}[x]$. If $p(\alpha) = 0$, then $p(\overline{\alpha}) = 0$. (Trivial if $\alpha \in \mathbb{R}$.)

Direct consequence:

(1) If $p(x) \in \mathbb{R}[x]$ is of **odd** degree, then $p(x)$ has at least one real root.

(2) Note that

$$(x-\alpha)(x-\overline{\alpha}) = x^2 - (\alpha+\overline{\alpha})x + \alpha\overline{\alpha} = x^2 - 2\,\mathrm{Re}(\alpha)x + |\alpha|^2.$$

Every $p(x) \in \mathbb{R}[x] \subseteq \mathbb{C}[x]$ can be factorized into a product of linear and quadratic factors with real coefficients:

$$\begin{aligned} p(x) &= (x-\beta_1)\cdots(x-\beta_r)(x-\alpha_1)(x-\overline{\alpha_1})\cdots(x-\alpha_s)(x-\overline{\alpha_s}) \\ &= \underbrace{(x-\beta_1)\cdots(x-\beta_r)}_{\text{real linear factors}}\underbrace{(x^2-2\,\mathrm{Re}(\alpha_1)x+|\alpha_1|^2)\cdots(x^2-2\,\mathrm{Re}(\alpha_s)x+|\alpha_s|^2)}_{\text{real quadratic factors}} \end{aligned}$$

where $\beta_1, \ldots \beta_r \in \mathbb{R}$ and $\alpha_1, \overline{\alpha_1}, \ldots, \alpha_s, \overline{\alpha_s} \in \mathbb{C}$ $\quad (\deg(p(x)) = r + 2s)$.

Therefore $p(\overline{\alpha}) = 0$.

Roots of Unity

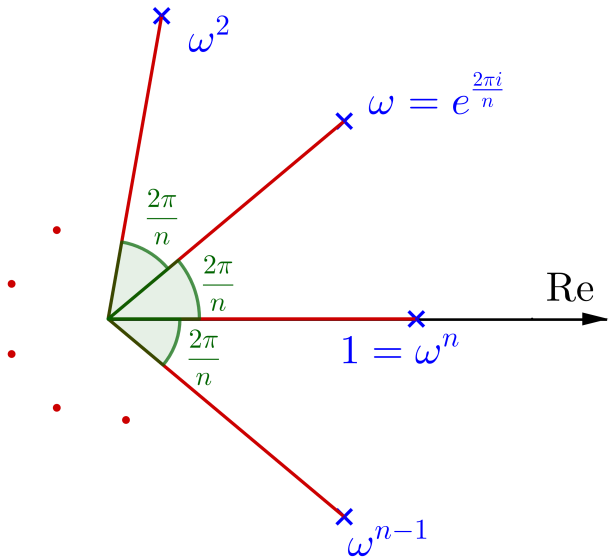
Factorizing $z^n - 1$ is equivalent to solving $z^n - 1 = 0$ (i.e. $z^n = 1$) $(*)$.

- Of course, 1 is a root of $(*)$. What about the remaining roots?

$$\text{Write } 1 = e^{2\pi i} = \left(e^{\frac{2\pi i}{n}}\right)^n.$$

- Let $\omega = e^{\frac{2\pi i}{n}} = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$.
  Then, $\omega^n = \left(e^{\frac{2\pi i}{n}}\right)^n = e^{2\pi i} = 1$. So ω is a root of $(*)$.
- Actually, ω^k is a root of $(*)$ for all $k \in \mathbb{Z}$ since $(\omega^k)^n = (\omega^n)^k = 1^k = 1$.
- Now, $1, \omega, \omega^2, \ldots, \omega^{n-1}$ are n distinct roots of $(*)$ (called the **n-th roots of unity**). Hence,

$$z^n - 1 = (z-1)(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}).$$

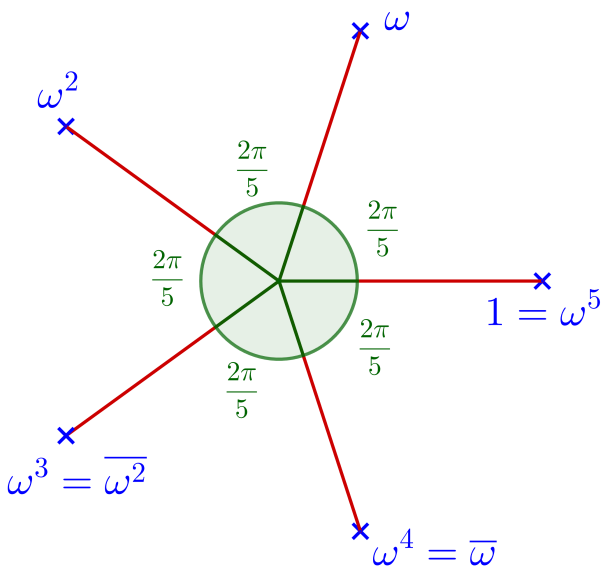


Example 6.9

Factorize $z^5 - 1$.

Let $\omega = e^{\frac{2\pi i}{5}}$. Then,

$$
\begin{aligned}
z^5 - 1 &= (z-1)(z-\omega)(z-\omega^2)(z-\omega^3)(z-\omega^4) && \text{(over } \mathbb{C}\text{)} \\
&= (z-1)(z-\omega)(z-\omega^2)(z-\overline{\omega^2})(z-\overline{\omega}) && (\overline{\omega} = \omega^4,\ \overline{\omega^2} = \omega^3) \\
&= (z-1)(z^2 - 2\operatorname{Re}(\omega)z + |\omega|^2)(z^2 - 2\operatorname{Re}(\omega^2)z + |\omega^2|^2) \\
&= (z-1)(z^2 - 2(\cos\frac{2\pi}{5})z + 1)(z^2 - 2(\cos\frac{4\pi}{5})z + 1) && \text{(over } \mathbb{R}\text{)}
\end{aligned}
$$

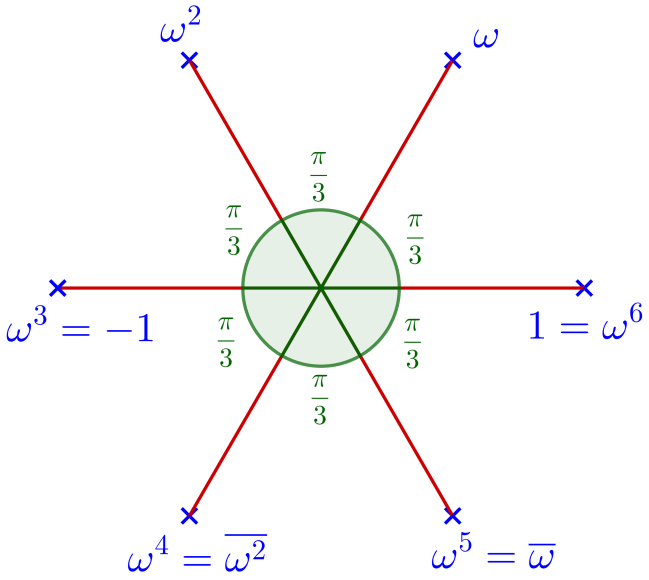


Example 6.10

Factorize $z^6 - 1$.

Let $\omega = e^{\frac{2\pi i}{6}} = e^{\frac{\pi i}{3}}$. Then,

$$\begin{aligned}
z^6 - 1 &= (z-1)(z-\omega)(z-\omega^2)(z-\omega^3)(z-\omega^4)(z-\omega^5) && \text{(over } \mathbb{C}\text{)} \\
&= (z-1)(z-\omega)(z-\omega^2)(z-\omega^3)(z-\overline{\omega^2})(z-\overline{\omega}) && (\overline{\omega^k} = \omega^{6-k}) \\
&= (z-1)(z+1)(z^2 - 2\operatorname{Re}(\omega)z + |\omega|^2)(z^2 - 2\operatorname{Re}(\omega^2)z + |\omega^2|^2) \\
&= (z-1)(z+1)(z^2 - 2(\cos\frac{\pi}{3})z + 1)(z^2 - 2(\cos\frac{2\pi}{3})z + 1) && \text{(over } \mathbb{R}\text{)}
\end{aligned}$$



In general,

$$z^n - 1 = \begin{cases} (z-1)\displaystyle\prod_{k=1}^{\frac{n-1}{2}} \left(z^2 - 2(\cos\frac{2k\pi}{n})z + 1\right) & \text{if } n \text{ is odd;} \\ (z-1)(z+1)\displaystyle\prod_{k=1}^{\frac{n}{2}-1} \left(z^2 - 2(\cos\frac{2k\pi}{n})z + 1\right) & \text{if } n \text{ is even.} \end{cases}$$

Exercise 6.7

Prove the formulas above.

Roots of Unity

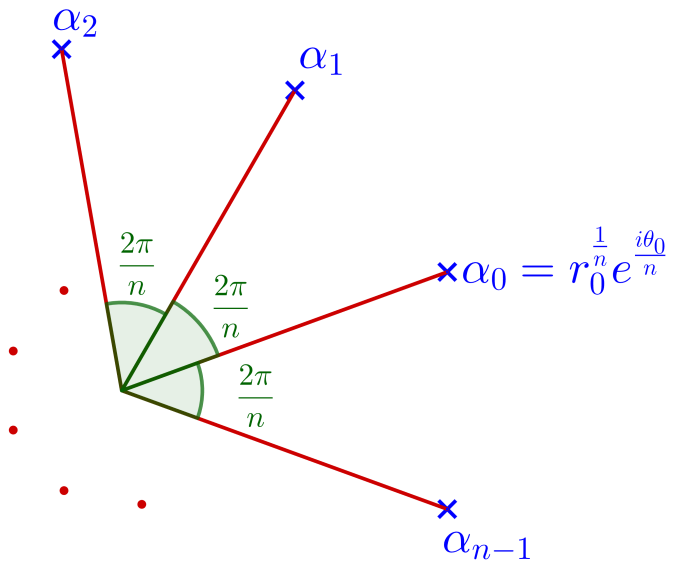
Similarly, we solve the equation

$$z^n = z_0, \qquad \text{where } n \in \mathbb{Z}^+ \text{ and } z_0 = r_0 e^{i\theta_0}.$$

Let $\omega = e^{\frac{2\pi i}{n}}$.
Then, $z^n = z_0$ has n distinct roots:

$$\alpha_k := r_0^{\frac{1}{n}} e^{\frac{i\theta_0}{n}} \omega^k = r_0^{\frac{1}{n}} e^{\frac{2k\pi+\theta_0}{n} i} = r_0^{\frac{1}{n}} \left[\cos\left(\frac{2k\pi+\theta_0}{n}\right) + i \sin\left(\frac{2k\pi+\theta_0}{n}\right)\right],$$

where $k = 0, 1, \ldots, n-1$.



Exercise 6.8

Factorize $z^5 + 1$ and $z^6 + 1$ over $\mathbb{R}$ and $\mathbb{C}$.

Example 6.11

Let n be a positive integer.

(a) Show that $(z - e^{i\theta})(z - e^{-i\theta}) = z^2 - 2z\cos\theta + 1$. Hence or otherwise, deduce that

$$z^{2n} + 1 = \prod_{k=1}^{n} \left(z^2 - 2z\cos\left(\tfrac{(2k-1)\pi}{2n}\right) + 1 \right).$$

(b) Show that, when n is even,

$$\cos\left(\tfrac{\pi}{2n}\right)\cos\left(\tfrac{3\pi}{2n}\right)\cos\left(\tfrac{5\pi}{2n}\right)\cdots\cos\left(\tfrac{(2n-1)\pi}{2n}\right) = (-1)^{\frac{1}{2}n}2^{1-n}.$$

(c) When n is odd, evaluate

$$\cos^2\left(\tfrac{\pi}{2n}\right)\cos^2\left(\tfrac{3\pi}{2n}\right)\cos^2\left(\tfrac{5\pi}{2n}\right)\cdots\cos^2\left(\tfrac{(n-2)\pi}{2n}\right).$$

Roots of Unity

(a) $(z - e^{i\theta})(z - e^{-i\theta}) = z^2 - (e^{i\theta} + e^{-i\theta})z + e^{i\theta}e^{-i\theta} = z^2 - 2z\cos\theta + 1.$
Need to show:

$$z^{2n} + 1 = \prod_{k=1}^{n} \left(z^2 - 2z\cos\left(\tfrac{(2k-1)\pi}{2n}\right) + 1 \right) .$$

The $(2n)$ roots of $z^{2n} + 1 = 0$ are

$$e^{\frac{i\pi}{2n}}, \qquad e^{\frac{i3\pi}{2n}}, \quad \ldots \quad , \quad e^{\frac{i(2n-1)\pi}{2n}}$$
$$e^{\frac{i(4n-1)\pi}{2n}}, \quad e^{\frac{i(4n-3)\pi}{2n}}, \quad \ldots \quad , \quad e^{\frac{i(2n+1)\pi}{2n}},$$

which can be written as

$$e^{i\frac{2k-1}{2n}\pi}, \quad e^{-i\frac{2k-1}{2n}\pi}, \qquad k = 1, 2, \ldots, n.$$

Hence,

$$\begin{aligned} z^{2n} + 1 &= \prod_{k=1}^{n} \left((z - e^{i\frac{2k-1}{2n}\pi})(z - e^{-i\frac{2k-1}{2n}\pi}) \right) \\ &= \prod_{k=1}^{n} \left(z^2 - 2z\cos\left(\tfrac{(2k-1)\pi}{2n}\right) + 1 \right) . \end{aligned}$$

(a)

$$z^{2n} + 1 = \prod_{k=1}^{n} \left(z^2 - 2z \cos\left(\tfrac{(2k-1)\pi}{2n} \right) + 1 \right) .$$

(b) Suppose n is even. Need to show

$$\cos\left(\tfrac{\pi}{2n}\right) \cos\left(\tfrac{3\pi}{2n}\right) \cos\left(\tfrac{5\pi}{2n}\right) \cdots \cos\left(\tfrac{(2n-1)\pi}{2n}\right) = (-1)^{\frac{1}{2}n} 2^{1-n} .$$

Putting $z = i$, we have

$$2 = (i)^{2n} + 1 = \prod_{k=1}^{n} \left(-1 - 2i \cos\left(\tfrac{(2k-1)\pi}{2n} \right) + 1 \right) ,$$

and so

$$\prod_{k=1}^{n} \cos\left(\tfrac{(2k-1)\pi}{2n} \right) = \frac{2}{(-2i)^n} = (-1)^{n/2} 2^{1-n}.$$

(c) Suppose n is odd.

When $k = \dfrac{n+1}{2}$, $\cos\left(\frac{(2\cdot\frac{n+1}{2}-1)\pi}{2n}\right) = \cos\frac{\pi}{2} = 0$. Moreover,

$$\cos\left(\tfrac{(2k-1)\pi}{2n}\right)\cos\left(\tfrac{(2n-2k+1)\pi}{2n}\right) = -\cos^2\left(\tfrac{(2k-1)\pi}{2n}\right) \qquad \text{for } k = 1, 2, \ldots, \tfrac{n-1}{2}.$$

Since $1 + z^{2n} = (1+z^2)(1 - z^2 + z^4 - \cdots + z^{2n-2})$, we have

$$1 - z^2 + z^4 - \cdots + z^{2n-2} = \prod_{\substack{k=1\\ k\neq\frac{n+1}{2}}}^{n} \left(z^2 - 2z\cos\left(\tfrac{(2k-1)\pi}{2n}\right) + 1\right).$$

Putting $z = i$, we have

$$n = \prod_{\substack{k=1\\ k\neq\frac{n+1}{2}}}^{n} \left(-2i\cos\left(\tfrac{(2k-1)\pi}{2n}\right)\right) = (-2i)^{n-1}\prod_{k=1}^{\frac{n-1}{2}}\left(-\cos^2\left(\tfrac{(2k-1)\pi}{2n}\right)\right).$$

Hence,

$$\begin{aligned}&\cos^2\left(\tfrac{\pi}{2n}\right)\cos^2\left(\tfrac{3\pi}{2n}\right)\cos^2\left(\tfrac{5\pi}{2n}\right)\cdots\cos^2\left(\tfrac{(n-2)\pi}{2n}\right)\\ &= n2^{1-n}(-1)^{\frac{n-1}{2}}(-1)^{\frac{n-1}{2}} = n2^{1-n}.\end{aligned}$$

Example 6.12

Let $n \in \mathbb{Z}^+$. Let $\omega = e^{\frac{2\pi i}{n}} = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$.

(a) Show that for any $m \in \mathbb{Z}$, $\omega^m, \omega^{m+1}, \ldots, \omega^{m+n-1}$ are n distinct roots of $z^n - 1 = 0$.

(b) Show that $1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0$.

(c) Evaluate $\omega \times \omega^2 \times \cdots \times \omega^{n-1}$.

(a) It suffices to show that $\omega^m, \omega^{m+1}, \ldots, \omega^{m+n-1}$ are distinct. Suppose not. Then there are $i, j \in \{0, 1, \ldots, n-1\}$ with $i < j$ such that $\omega^{m+i} = \omega^{m+j}$. So $\omega^i = \omega^{-m} \cdot \omega^{m+i} = \omega^{-m} \cdot \omega^{m+j} = \omega^j$. This contradicts the fact that $\omega, \omega, \ldots, \omega^{n-1}$ are distinct.

(b) Since $(\omega - 1)(1 + \omega + \omega^2 + \cdots + \omega^{n-1}) = \omega^n - 1 = 0$ and $\omega \neq 1$, we have $1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0$.

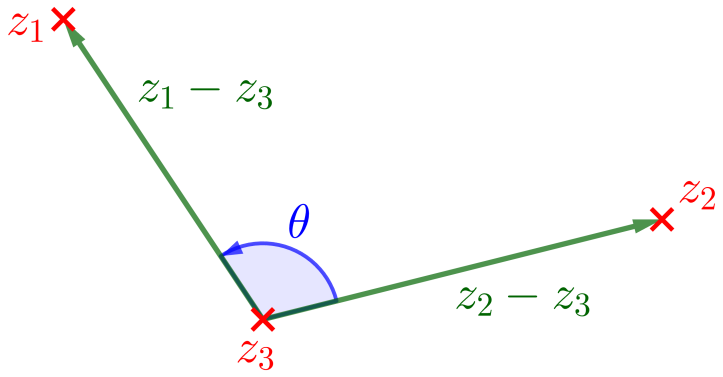
(c) Recall that

$$z^n - 1 = (z-1)(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}).$$

Putting $z = 0$, we have $-1 = (-1)(-\omega)(-\omega^2)\cdots(-\omega^{n-1})$. Therefore $\omega \times \omega^2 \times \cdots \times \omega^{n-1} = (-1)^{n-1}$.

Geometry Involving Complex Numbers

Let $z_1, z_2, z_3 \in \mathbb{C}$ be distinct points.
Write $\dfrac{z_1 - z_3}{z_2 - z_3} = re^{i\theta}$ (polar form).



Then

$$z_1 - z_3 = \frac{z_1 - z_3}{z_2 - z_3} \cdot (z_2 - z_3) = re^{i\theta}(z_2 - z_1).$$

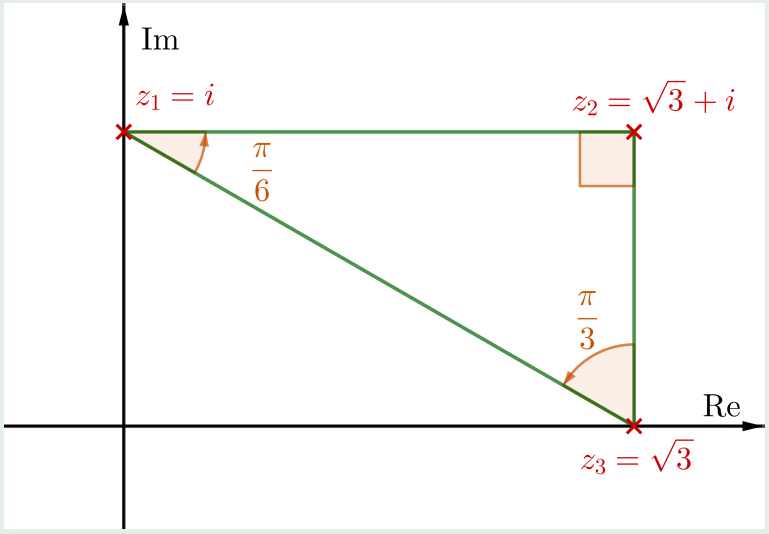
Therefore, $z_1 - z_3$ can be obtained by rotating $z_2 - z_3$ by θ and then scaling by r.

$$r = \left|\frac{z_1 - z_3}{z_2 - z_3}\right| = \frac{|z_1 - z_3|}{|z_2 - z_3|} \qquad \text{ratio of two sides}$$

$$\theta = \arg\left(\frac{z_1 - z_3}{z_2 - z_3}\right) = \angle z_2 z_3 z_1 \qquad \text{angle between two sides (with orientation)}$$

Example 6.13

Let $z_1 = i$, $z_2 = \sqrt{3} + i$ and $z_3 = \sqrt{3}$.



Then

$$\frac{z_1 - z_3}{z_2 - z_3} = 2e^{\frac{\pi i}{3}}$$

$$\frac{z_2 - z_1}{z_3 - z_1} = \frac{\sqrt{3}}{2} e^{\frac{\pi i}{6}}$$

$$\frac{z_3 - z_2}{z_1 - z_2} = \frac{1}{\sqrt{3}} e^{\frac{\pi i}{2}}.$$

Proposition 6.9

Three points $z_1, z_2, z_3 \in \mathbb{C}$ are collinear if and only if $\begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ \overline{z_1} & \overline{z_2} & \overline{z_3} \end{vmatrix} = 0.$

Recall the definition of determinant:
The determinant of a 2×2 matrix is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

and the determinant of a 3×3 matrix is

$$\begin{aligned} \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} &= aei + bfg + cdh - ceg - bdi - afh \\ &= a \cdot \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \cdot \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \cdot \begin{vmatrix} d & e \\ g & h \end{vmatrix}. \end{aligned}$$

Proposition 6.9

Three points $z_1, z_2, z_3 \in \mathbb{C}$ are collinear if and only if $\begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ \overline{z_1} & \overline{z_2} & \overline{z_3} \end{vmatrix} = 0.$

Proof.

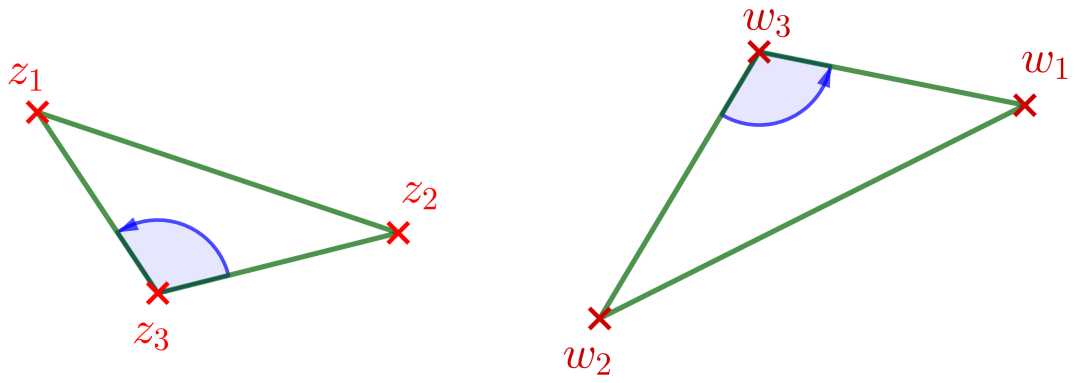
$$\begin{aligned}
& z_1, z_2, z_3 \text{ are collinear} \\
\iff \quad & \operatorname{Arg}\left(\frac{z_1 - z_3}{z_2 - z_3}\right) = 0 \quad \text{or} \quad \pi \\
\iff \quad & \frac{z_1 - z_3}{z_2 - z_3} \in \mathbb{R} \\
\iff \quad & \frac{z_1 - z_3}{z_2 - z_3} = \frac{\overline{z_1} - \overline{z_3}}{\overline{z_2} - \overline{z_3}} \\
\iff \quad & (z_1 - z_3)(\overline{z_2} - \overline{z_3}) - (z_2 - z_3)(\overline{z_1} - \overline{z_3}) = 0 \\
\iff \quad & (z_2\overline{z_3} - \overline{z_2}z_3) - (z_1\overline{z_3} - \overline{z_1}z_3) + (z_1\overline{z_2} - \overline{z_1}z_2) = 0 \\
\iff \quad & \begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ \overline{z_1} & \overline{z_2} & \overline{z_3} \end{vmatrix} = 0.
\end{aligned}$$

Exercise 6.9

Show that $\dfrac{z_1 - z_3}{z_2 - z_3} \in \mathbb{R} \iff \dfrac{z_2 - z_1}{z_3 - z_1} \in \mathbb{R} \iff \dfrac{z_3 - z_2}{z_1 - z_2} \in \mathbb{R}$.

Geometry Involving Complex Numbers

Consider two triangles $\triangle z_1 z_2 z_3$, $\triangle w_1 w_2 w_3$ with vertices $z_1, z_2, z_3, w_1, w_2, w_3 \in \mathbb{C}$.



Then,

$$\frac{z_1 - z_3}{z_2 - z_3} = \frac{w_1 - w_3}{w_2 - w_3}$$

$$\iff \quad \underbrace{\frac{|z_1 - z_3|}{|z_2 - z_3|} = \frac{|w_1 - w_3|}{|w_2 - w_3|}}_{\text{same as}} \quad \text{and} \quad \underbrace{\arg\left(\frac{z_1 - z_3}{z_2 - z_3}\right) = \arg\left(\frac{w_1 - w_3}{w_2 - w_3}\right)}_{\substack{\text{(Not just the magnitude} \\ \text{but also the orientation)}}}$$

$$\frac{|z_1 - z_3|}{|w_1 - w_3|} = \frac{|z_2 - z_3|}{|w_2 - w_3|}$$

$$\iff \quad \triangle z_1 z_2 z_3 \sim \triangle w_1 w_2 w_3 \quad \text{(with the same orientation)}$$

(ratio of two sides, inc. $\angle$)

Proposition 6.10

Let $z_1, z_2, z_3, w_1, w_2, w_3 \in \mathbb{C}$. *Then*

$$\triangle z_1 z_2 z_3 \underset{\substack{\uparrow \\ \textit{(same orientation)}}}{\sim} \triangle w_1 w_2 w_3 \iff \frac{z_1 - z_3}{z_2 - z_3} = \frac{w_1 - w_3}{w_2 - w_3} \iff \begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = 0.$$

Corollary 6.11

$$\triangle z_1 z_2 z_3 \textit{ is equilateral} \iff \triangle z_1 z_2 z_3 \sim \triangle z_2 z_3 z_1 \iff \begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ z_2 & z_3 & z_1 \end{vmatrix} = 0.$$

The last condition is equivalent to

$$z_1^2 + z_2^2 + z_3^2 - z_1 z_2 - z_2 z_3 - z_3 z_1 = 0.$$

Example 6.14

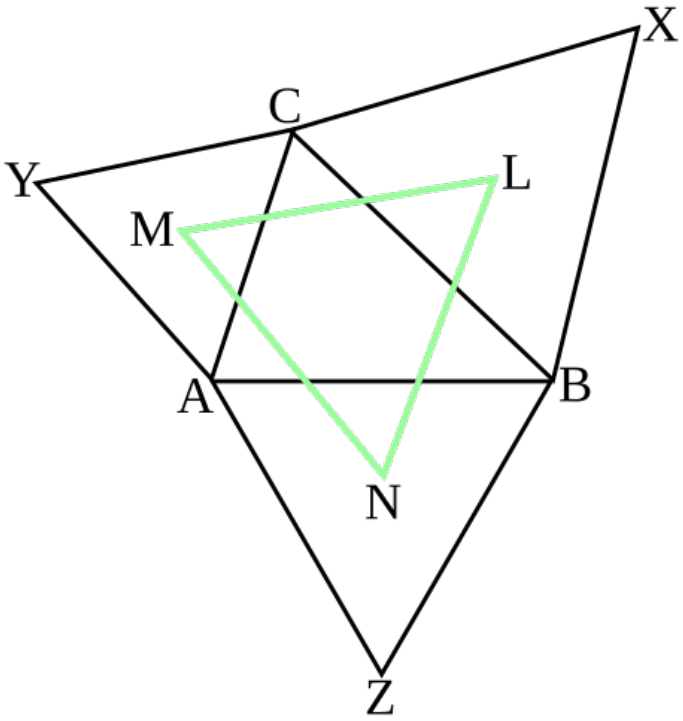
Let $z_1, z_2, z_3 \in \mathbb{C}$ are the vertices of a triangle in the counter-clockwise orientation. Show that $\triangle z_1 z_2 z_3$ is an equilateral triangle if and only if $z_1 + z_2\omega + z_3\omega^2 = 0$, where $\omega = e^{\frac{2\pi i}{3}}$ is a cube root of unity.

Let $w_1 = 1, w_2 = \omega, w_3 = \omega^2$.
Note that $\triangle z_1 z_2 z_3$ is equilateral if and only if $\triangle z_1 z_2 z_3 \sim \triangle w_1 w_2 w_3$.
By Proposition 6.10, this is equivalent to

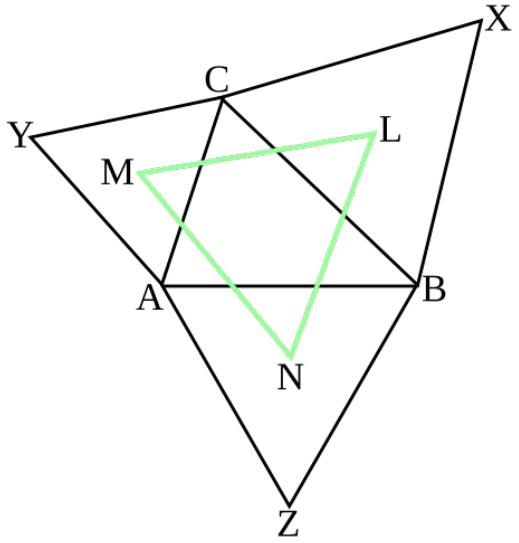
$$\begin{aligned}
& \frac{z_1 - z_3}{z_2 - z_3} = \frac{w_1 - w_3}{w_2 - w_3} = \frac{1 - \omega^2}{\omega - \omega^2} = \frac{1 + \omega}{\omega} \\
\iff \quad & z_1\omega - z_3\omega = z_2(1 + \omega) - z_3(1 + \omega) \\
\iff \quad & z_1\omega + z_2\omega^2 + z_3 = 0 \\
\iff \quad & z_1 + z_2\omega + z_3\omega^2 = \omega^2(z_1\omega + z_2\omega^2 + z_3) = 0.
\end{aligned}$$

Exercise 6.10 (Napoleon's Theorem)

Given $\triangle ABC$. Draw equilateral triangles $\triangle BXC, \triangle CYA, \triangle AZB$ on the opposite side of BC, CA, AB, respectively. Let L, M, N be the centroids of $\triangle BXC, \triangle CYA, \triangle AZB$, respectively. Then $\triangle LMN$ is equilateral.



Napoleon's theorem
https://en.wikipedia.org/wiki/Napoleon%27s_theorem



Denote the points $A, B, C, X, Y, Z, M, L, N$ by $a, b, c, x, y, z, m, l, n \in \mathbb{C}$. Since $\triangle BXC, \triangle CYA, \triangle AZB$ are equilateral and are in the counter-clockwise orientation, we have

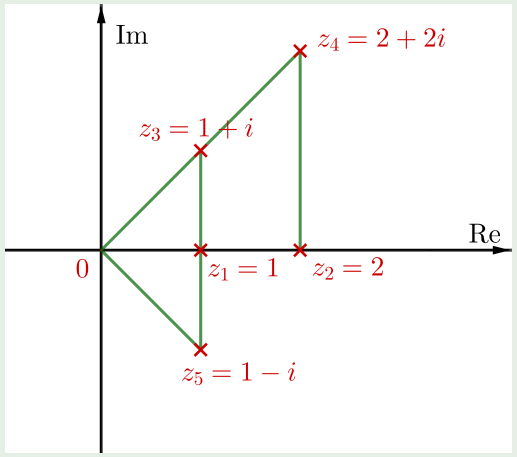
$$x + c\omega + b\omega^2 = 0, \quad y + a\omega + c\omega^2 = 0, \quad z + b\omega + a\omega^2 = 0.$$

Since $\omega^3 = 1$, we have

$$\begin{aligned} l + m\omega + n\omega^2 &= \frac{b+x+c}{3} + \frac{c+y+a}{3}\omega + \frac{a+z+b}{3}\omega^2 \\ &= \frac{1}{3}\left[(x + c\omega + b\omega^2) + \omega(y + a\omega + c\omega^2) + \omega^2(z + b\omega + a\omega^2)\right] \\ &= 0. \end{aligned}$$

Example 6.15

Let $z_0 = 0, z_1 = 1, z_2 = 2, z_3 = 1 + i, z_4 = 2 + 2i, z_5 = 1 - i$.



Then

$$\frac{z_1 - 0}{z_3 - 0} = \frac{1}{2} - \frac{1}{2}i = \frac{z_2 - 0}{z_4 - 0}$$
$$\frac{z_1 - 0}{z_5 - 0} = \frac{1}{2} + \frac{1}{2}i = \overline{\left(\frac{z_1 - 0}{z_3 - 0}\right)}$$

Note that $\triangle z_0 z_2 z_4 \sim \triangle z_0 z_1 z_3 \overset{\text{rev}}{\sim} \triangle z_0 z_1 z_5$.

$\uparrow$

(orientation reversed)

Proposition 6.12

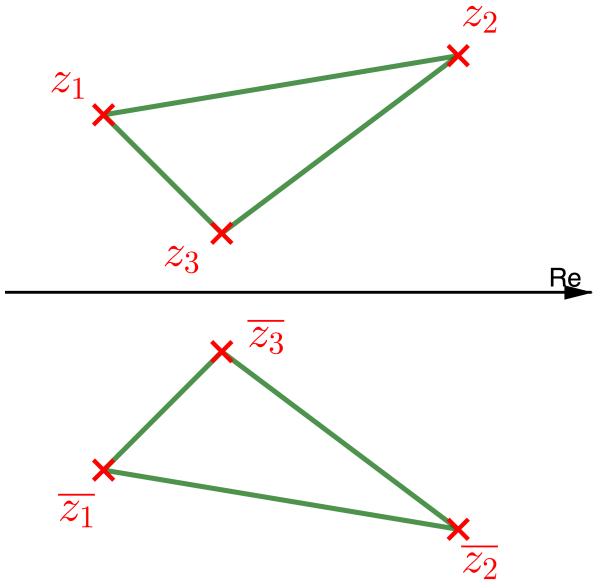
Let $z_1, z_2, z_3, w_1, w_2, w_3 \in \mathbb{C}$. *Then*

$$\triangle z_1 z_2 z_3 \underset{\uparrow}{\overset{\text{rev}}{\sim}} \triangle w_1 w_2 w_3 \iff \frac{z_1 - z_3}{z_2 - z_3} = \frac{\overline{w_1} - \overline{w_3}}{\overline{w_2} - \overline{w_3}} \iff \begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ \overline{w_1} & \overline{w_2} & \overline{w_3} \end{vmatrix} = 0.$$

(orientation reversed)

Another interpretation of Proposition 6.9:

$$z_1, z_2, z_3 \text{ are collinear} \iff \triangle z_1 z_2 z_3 \sim \triangle \overline{z_1}\,\overline{z_2}\,\overline{z_3} \iff \begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ \overline{z_1} & \overline{z_2} & \overline{z_3} \end{vmatrix} = 0.$$

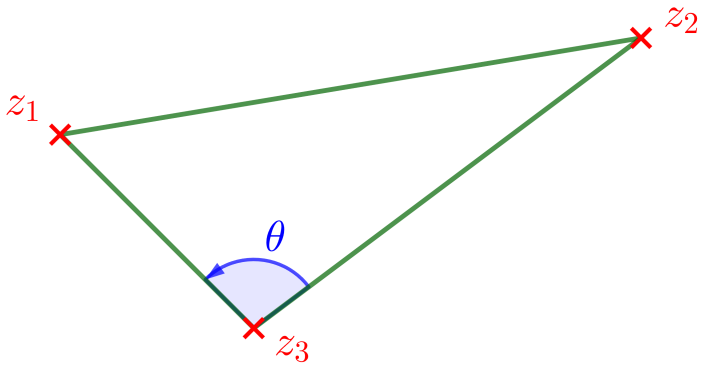


Geometry Involving Complex Numbers

Signed area of $\triangle z_1 z_2 z_3$

$$= \frac{1}{2}|z_1 - z_3||z_2 - z_3| \sin\theta$$

$$= \frac{1}{2}|z_2 - z_3|^2 \left|\frac{z_1 - z_3}{z_2 - z_3}\right| \sin\theta$$



$$\left|\frac{z_1 - z_3}{z_2 - z_3}\right| \sin\theta = \operatorname{Im}\left(\frac{z_1 - z_3}{z_2 - z_3}\right) = \frac{1}{2i}\left(\frac{z_1 - z_3}{z_2 - z_3} - \frac{\overline{z_1} - \overline{z_3}}{\overline{z_2} - \overline{z_3}}\right)$$

$$= \frac{1}{2i|z_2 - z_3|^2} \begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ \overline{z_1} & \overline{z_2} & \overline{z_3} \end{vmatrix}.$$

So signed area of $\triangle z_1 z_2 z_3 = \dfrac{1}{4i} \begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ \overline{z_1} & \overline{z_2} & \overline{z_3} \end{vmatrix}.$

Another interpretation of Proposition 6.9:

z_1, z_2, z_3 collinear $\iff$ signed area of $\triangle z_1 z_2 z_3 = 0 \iff \begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ \overline{z_1} & \overline{z_2} & \overline{z_3} \end{vmatrix} = 0$

Definition 6.5

Let $z_1, z_2, z_3, z_4 \in \mathbb{C}$ be four distinct points. The cross ratio $(z_1, z_2; z_3, z_4)$ is defined as

$$(z_1, z_2; z_3, z_4) = \left(\frac{z_1 - z_3}{z_2 - z_3}\right) \Big/ \left(\frac{z_1 - z_4}{z_2 - z_4}\right) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_2 - z_3)(z_1 - z_4)}.$$

Exercise 6.11

Let $z_1, z_2, z_3, z_4 \in \mathbb{C}$ be four distinct points. Show that

(i) $(z_1, z_2; z_3, z_4) = (z_3, z_4; z_1, z_2)$

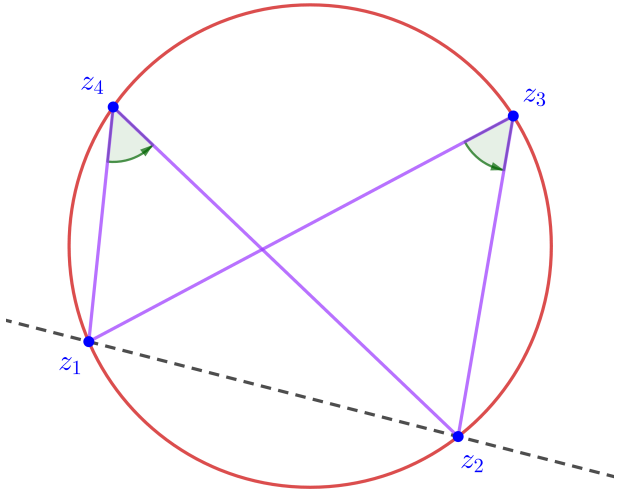
(ii) $(z_1, z_2; z_3, z_4) = 1/(z_1, z_2; z_4, z_3)$

(iii) $(z_1, z_2; z_3, z_4) = 1 - (z_1, z_3; z_2, z_4)$

Geometry Involving Complex Numbers

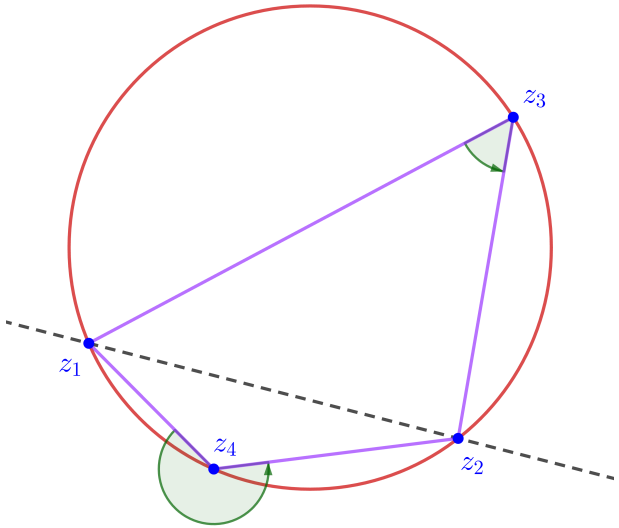
Let $z_1, z_2, z_3, z_4 \in \mathbb{C}$ be four distinct points.
Let $\mathcal{C}$ be the circle passing through z_1, z_2, z_3.
Then, z_4 lies on $\mathcal{C}$ if and only if

$$
\begin{aligned}
(1) \qquad & \angle z_1 z_3 z_2 = \angle z_1 z_4 z_2 \\
\iff \quad & \arg\left(\frac{z_2 - z_3}{z_1 - z_3}\right) = \arg\left(\frac{z_2 - z_4}{z_1 - z_4}\right) \\
\iff \quad & \arg\left(\frac{z_2 - z_3}{z_1 - z_3} \Big/ \frac{z_2 - z_4}{z_1 - z_4}\right) = 0 \\
\iff \quad & 1/(z_1, z_2; z_3, z_4) \in \mathbb{R}
\end{aligned}
$$



or

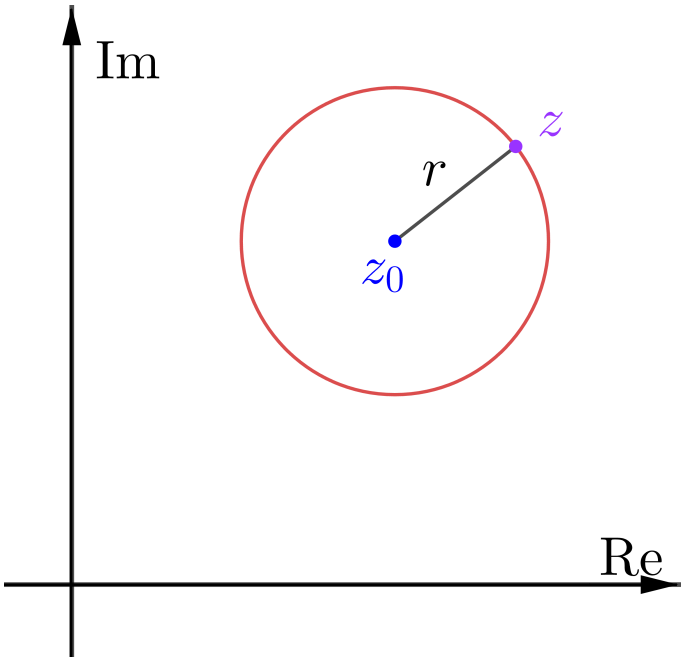
$$
\begin{aligned}
(2) \qquad & \angle z_1 z_3 z_2 - \angle z_1 z_4 z_2 = -\pi \\
\iff \quad & \arg\left(\frac{z_2 - z_3}{z_1 - z_3}\right) - \arg\left(\frac{z_2 - z_4}{z_1 - z_4}\right) = -\pi \\
\iff \quad & \arg\left(\frac{z_2 - z_3}{z_1 - z_3} \Big/ \frac{z_2 - z_4}{z_1 - z_4}\right) = -\pi \\
\iff \quad & 1/(z_1, z_2; z_3, z_4) \in \mathbb{R}
\end{aligned}
$$



Hence, z_1, z_2, z_3, z_4 are concyclic if and only if $(z_1, z_2; z_3, z_4) \in \mathbb{R}$.

(1) Circle centered at z_0 with radius $r > 0$:

$$|z - z_0| = r$$

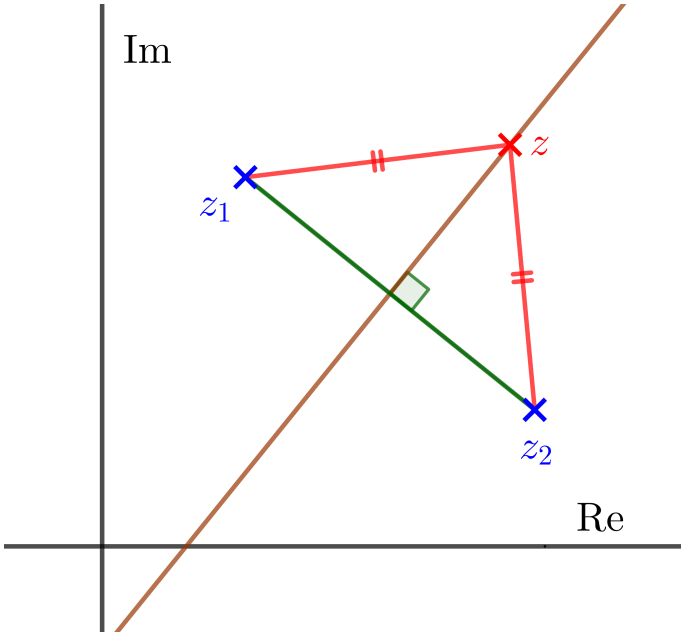


The equation can also be written as

$$\begin{aligned} |z - z_0|^2 &= r^2 \\ (z - z_0)(\overline{z} - \overline{z_0}) &= r^2 \\ z\overline{z} - \overline{z_0}z - z_0\overline{z} + (|z_0|^2 - r^2) &= 0. \end{aligned}$$

(2) Perpendicular bisector of the line segment joining z_1 and z_2 ($z_1 \neq z_2$):

$$|z - z_1| = |z - z_2|$$

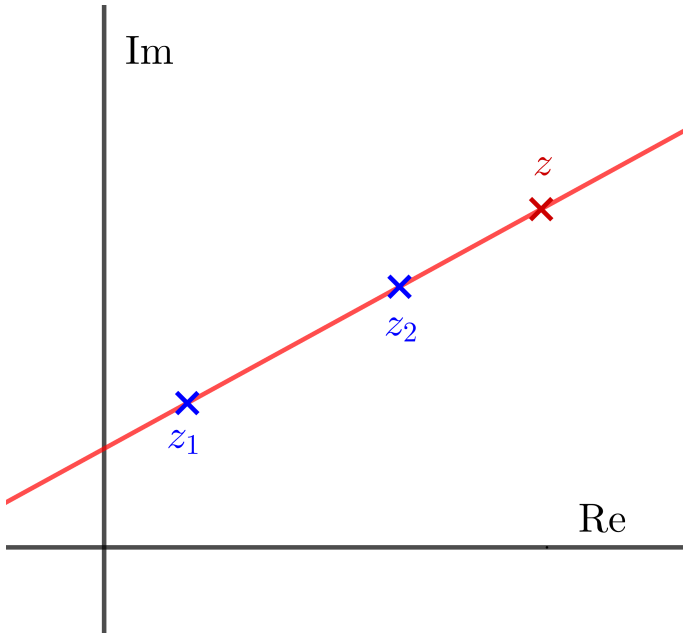


The equation can also be written as

$$\begin{aligned} |z - z_1|^2 &= |z - z_2|^2 \\ (z - z_1)(\overline{z} - \overline{z_1}) &= (z - z_2)(\overline{z} - \overline{z_2}) \\ \overline{(z_1 - z_2)}z + (z_1 - z_2)\overline{z} + (|z_2|^2 - |z_1|^2) &= 0. \end{aligned}$$

(3) Line joining z_1 and z_2 ($z_1 \neq z_2$):

$$z_1, z_2, z \text{ are collinear} \iff \frac{z - z_2}{z_1 - z_2} \in \mathbb{R} \iff \begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z \\ \overline{z_1} & \overline{z_2} & \overline{z} \end{vmatrix} = 0$$



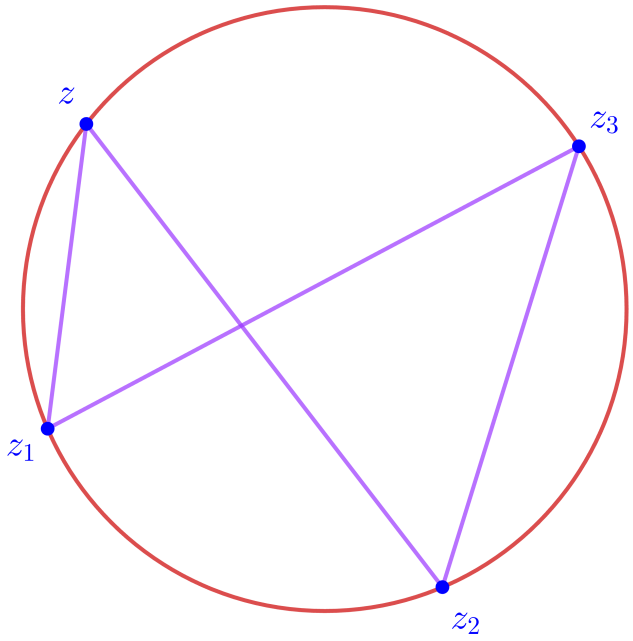
The equation can also be written as

$$\frac{z - z_2}{z_1 - z_2} - \frac{\overline{z} - \overline{z_2}}{\overline{z_1} - \overline{z_2}} = 0$$

$$(\overline{z_1} - \overline{z_2})(z - z_2) - (z_1 - z_2)(\overline{z} - \overline{z_2}) = 0.$$

(4) Circle passing through noncollinear points z_1, z_2, z_3:

$$(z_1, z_2; z_3, z) = \frac{(z_1 - z_3)(z_2 - z)}{(z_2 - z_3)(z_1 - z)} \in \mathbb{R}.$$



The equation can also be written as

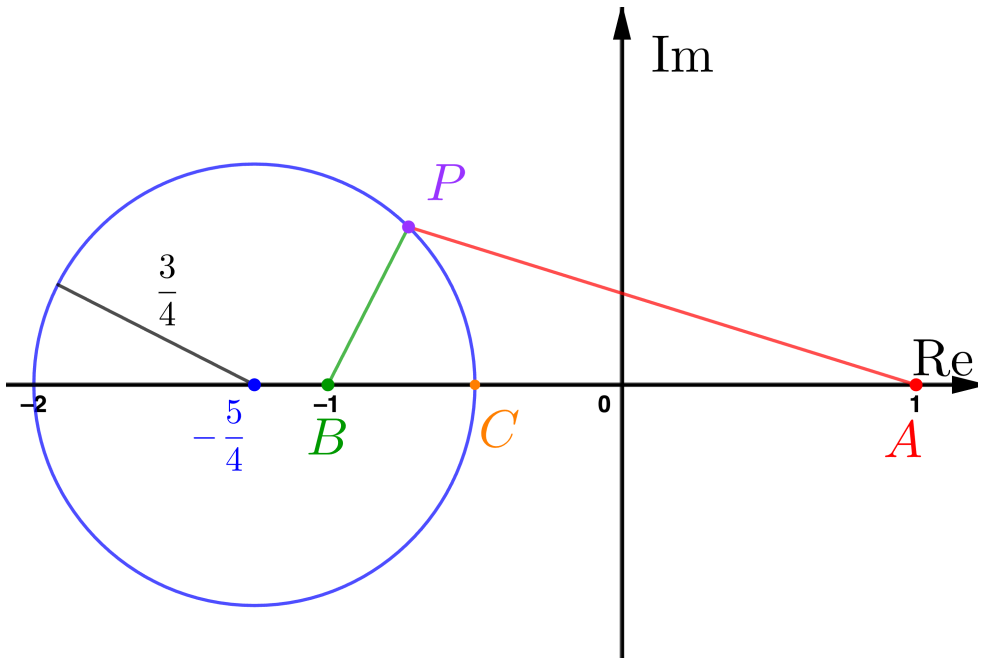
$$\frac{(z_1 - z_3)(z_2 - z)}{(z_2 - z_3)(z_1 - z)} - \frac{(\overline{z_1} - \overline{z_3})(\overline{z_2} - \overline{z})}{(\overline{z_2} - \overline{z_3})(\overline{z_1} - \overline{z})} = 0$$

Example 6.16

Show that $|z-1| = 3|z+1|$ represents a circle.

$$
\begin{aligned}
|z-1|^2 &= 9|z+1|^2 \\
(z-1)(\bar{z}-1) &= 9(z+1)(\bar{z}+1) \\
z\bar{z} - z - \bar{z} + 1 &= 9(z\bar{z} + z + \bar{z} + 1) \\
8z\bar{z} + 10z + 10\bar{z} + 8 &= 0 \\
z\bar{z} + \frac{5}{4}z + \frac{5}{4}\bar{z} + 1 &= 0 \\
(z + \frac{5}{4})(\bar{z} + \frac{5}{4}) - \frac{9}{16} &= 0 \\
|z + \frac{5}{4}| &= \frac{3}{4}
\end{aligned}
$$

So the equation represents a circle centered at $-\frac{5}{4}$ with radius $\frac{3}{4}$.



Apollonius's definition of a circle:

$$\frac{|AP|}{|BP|} = \frac{3}{1}.$$

Example 6.17

(a) If $|1+z| = |2-z|$, find $\mathrm{Re}(z)$.

(b) Find all $z \in \mathbb{C}$ such that $\begin{cases} |z|^2 - z - \bar{z} + i(z - \bar{z}) = \frac{1}{2} \\ |1+z| = |2-z| \end{cases}$

(a)

$$\begin{aligned} |1+z|^2 &= |2-z|^2 \\ (1+z)(1+\bar{z}) &= (2-z)(2-\bar{z}) \\ 1 + z + \bar{z} + z\bar{z} &= 4 - z - \bar{z} + z\bar{z} \\ z + \bar{z} &= 1 \\ \mathrm{Re}(z) &= \frac{1}{2} \quad \text{(it defines a vertical line)} \end{aligned}$$

(b)

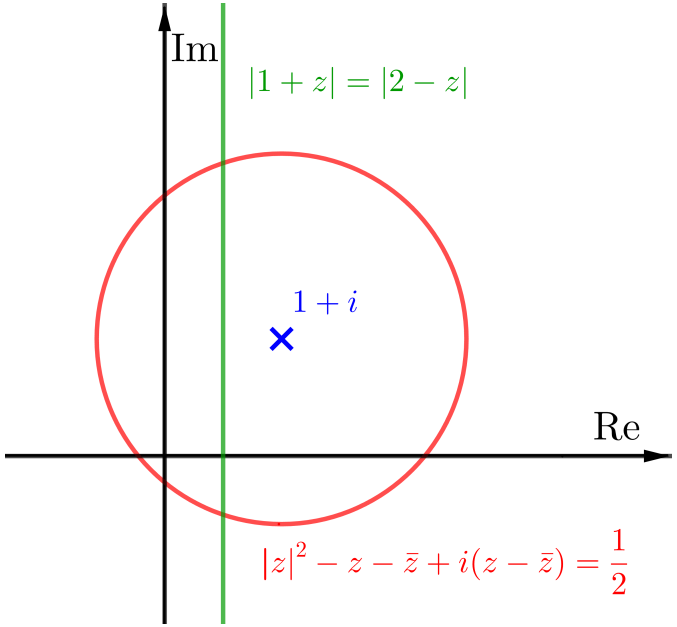
$$
\begin{aligned}
|z|^2 - z - \bar{z} + i(z - \bar{z}) &= \frac{1}{2} \\
z\bar{z} - (1-i)z - (1+i)\bar{z} &= \frac{1}{2} \\
[z - (1+i)]\left[\bar{z} - \overline{(1+i)}\right] &= \frac{1}{2} + |1+i|^2 \\
|z - (1+i)|^2 &= \frac{5}{2} \quad \text{(it defines a circle)}
\end{aligned}
$$

Put $z = \frac{1}{2} + iy$ where $y \in \mathbb{R}$. Then

$$(\frac{1}{2})^2 + (y-1)^2 = \frac{5}{2}$$

$$y = \frac{5}{2} \quad \text{or} \quad y = -\frac{1}{2}$$

So $z = \frac{1}{2} + \frac{5}{2}i$ or $z = \frac{1}{2} - \frac{1}{2}i$.



Example 6.18

Sketch the set $\mathrm{Im}(\frac{1}{z}) > 1$ on the complex plane.

For $z = x + iy \neq 0$,

$$\frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{\bar{z}}{|z|^2} = \frac{x - iy}{x^2 + y^2}.$$

The inequality $\mathrm{Im}(\frac{1}{z}) > 1$ becomes

$$\frac{-y}{x^2 + y^2} > 1,$$

that is

$$x^2 + y^2 + y < 0$$
$$(x - 0)^2 + (y + \frac{1}{2})^2 < (\frac{1}{2})^2.$$

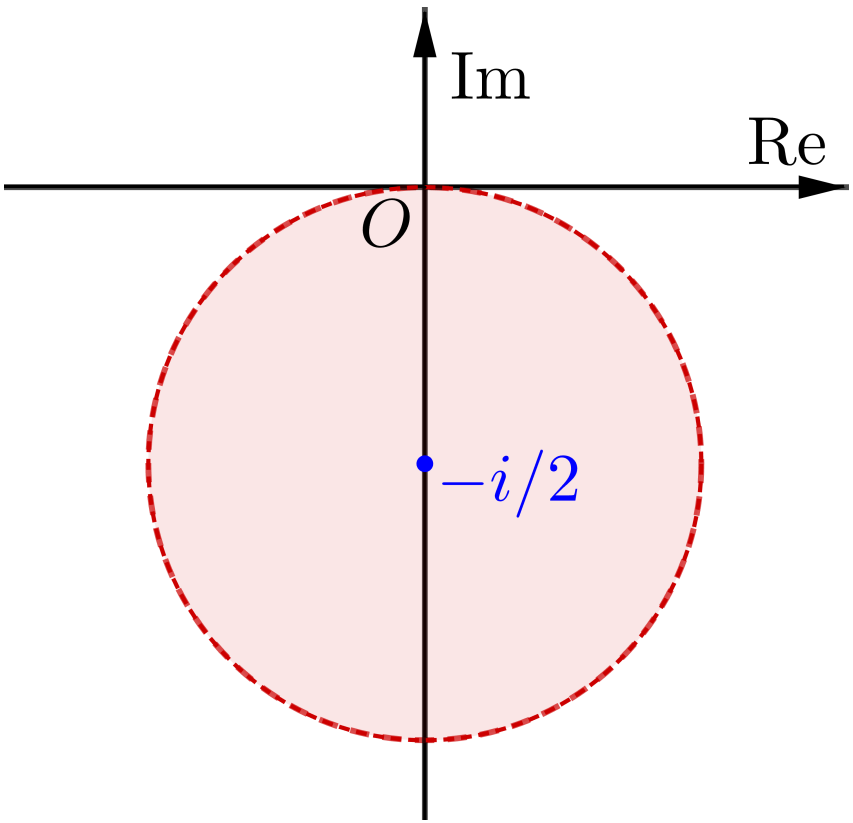
So the inequality represents the region interior to the circle centred at $z = -i/2$ and with radius $1/2$.

Example 6.18

Sketch the set $\mathrm{Im}(\frac{1}{z}) > 1$ on the complex plane.

For $z = x + iy$,

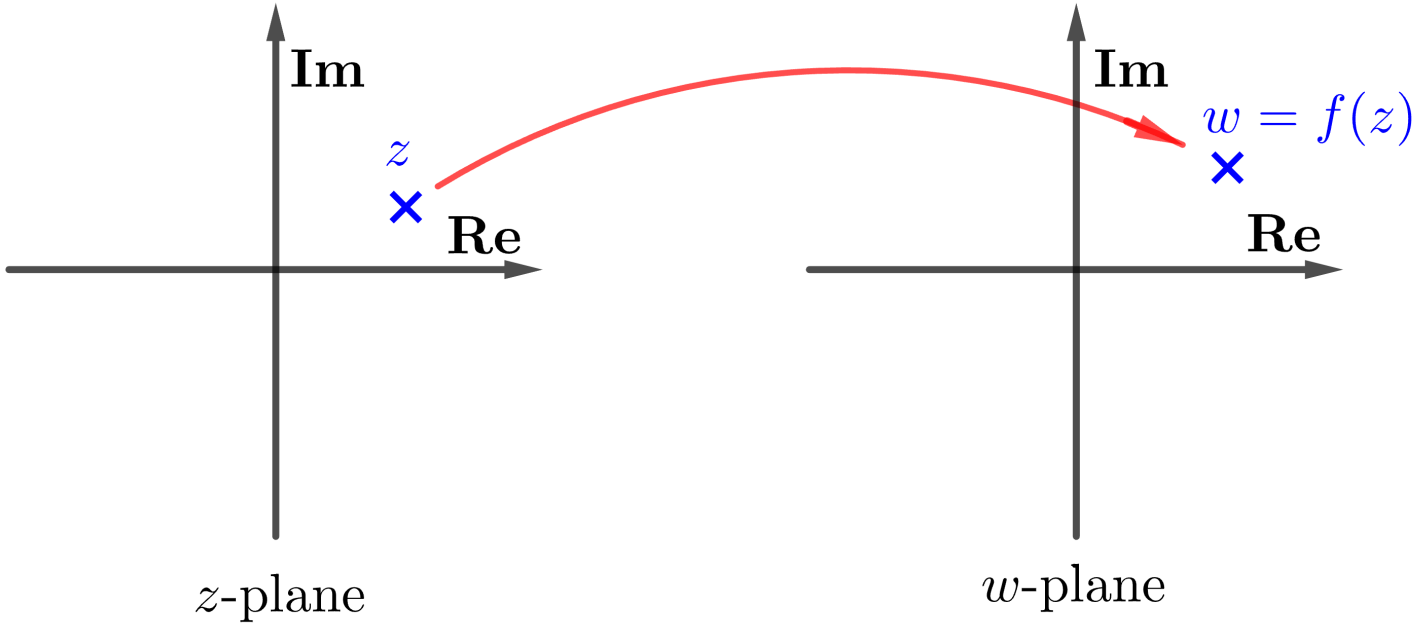
$$\mathrm{Im}(\frac{1}{z}) > 1 \iff (x-0)^2 + (y+\frac{1}{2})^2 < (\frac{1}{2})^2.$$



Complex Transformation

Definition 6.6

A **complex transformation** is a function from $\mathbb{C}$ to itself.



GeoGebra Illustration: Complex Transformation

Complex Transformation (Optional)

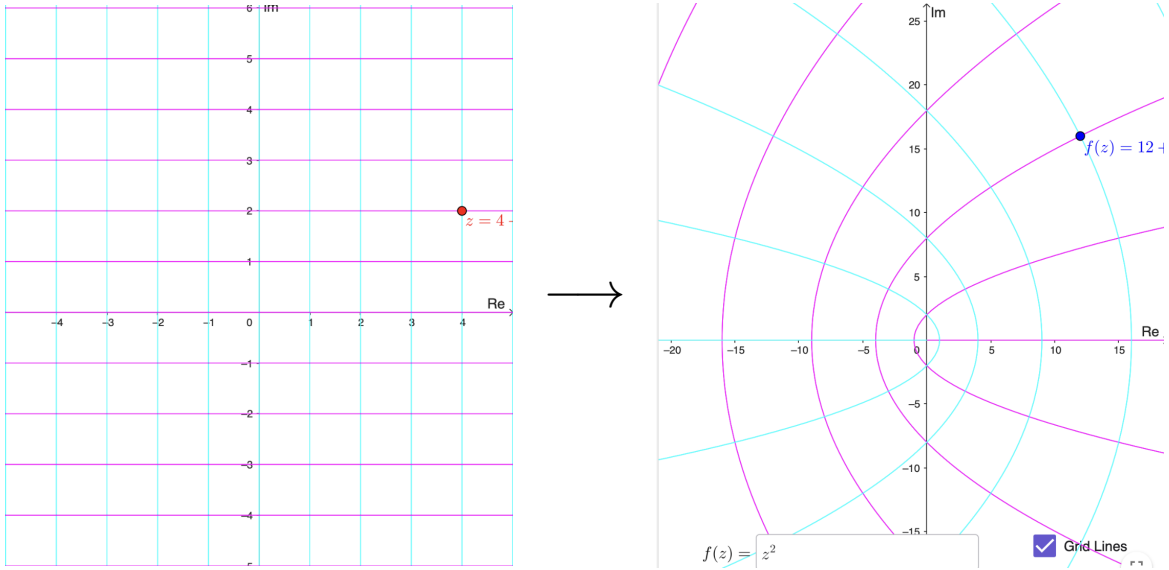
Example 6.19

Let $f : \mathbb{C} \to \mathbb{C}$ be a complex transformation defined by $w = f(z) = z^2$.
Write $z = x + iy$ and $w = u + iv$, where $x, y, u, v \in \mathbb{R}$.
We have

$$u + iv = w = z^2 = (x + iy)^2 = (x^2 - y^2) + i(2xy),$$

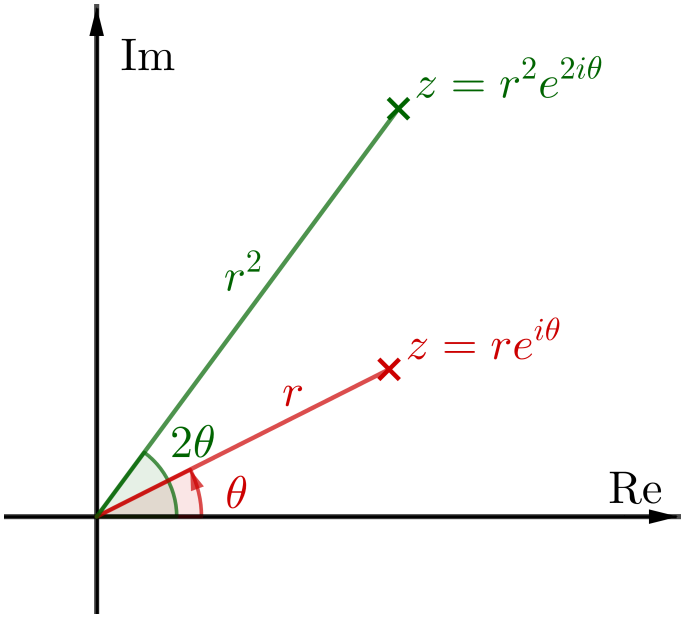
that is

$$\begin{cases} u(x, y) = x^2 - y^2 \\ v(x, y) = 2xy. \end{cases}$$



Complex Transformation (Optional)

We may also draw z and w on the same plane to compare them:
Write $z = re^{i\theta}$. Then $w = f(z) = z^2 = r^2 e^{2i\theta}$.

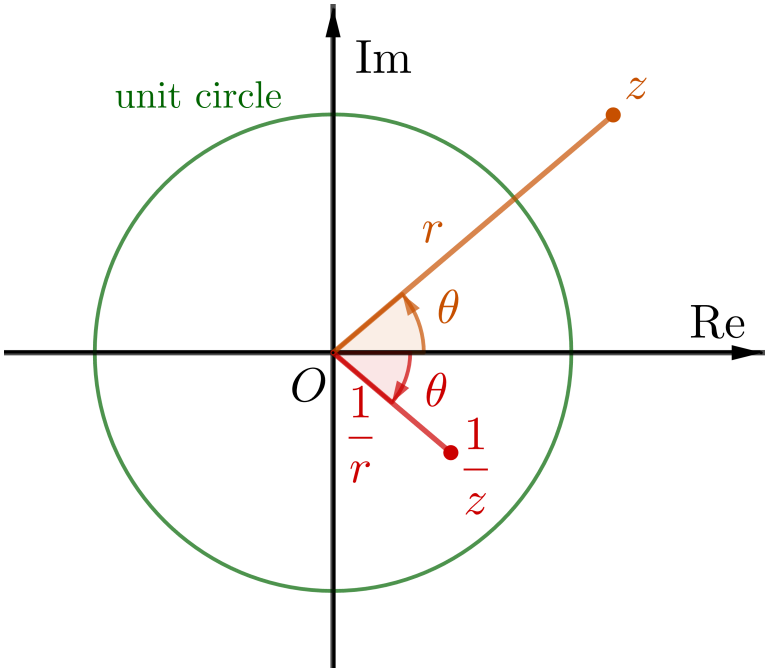


Complex Transformation (Optional)

Some special transformation:

$f(z) = rz$	$r \in \mathbb{R}\backslash\{0\}$	(scaling)
$f(z) = z + a$	$a \in \mathbb{C}$	(translation)
$f(z) = e^{i\theta} z$	$\theta \in \mathbb{R}$	(rotation)
$f(z) = \dfrac{1}{z}$	$(z \neq 0)$	(inversion)
$f(z) = \bar{z}$		(complex conjugation/reflection along the real axis)

Inversion:



Complex Transformation (Optional)

Exercise 6.12

Show that the distance between any two points is preserved under a translation, a rotation or the complex conjugation, i.e.

$$|f(z_1) - f(z_2)| = |z_1 - z_2| \qquad \text{for all } z_1, z_2 \in \mathbb{C}$$

if f is one of the above transformation.

Complex Transformation (Optional)

Exercise 6.13

Let $z_1, z_2, z_3 \in \mathbb{C}$ be distinct points. Show that the 3-point ratio $\dfrac{z_1 - z_3}{z_2 - z_3}$ is preserved under a translation, a rotation or a scaling, i.e.

$$\frac{f(z_1) - f(z_3)}{f(z_2) - f(z_3)} = \frac{z_1 - z_3}{z_2 - z_3} \qquad \text{for all distinct points } z_1, z_2, z_3 \in \mathbb{C}$$

if f is one of the above transformation.

Direct cosequence:

- $\triangle z_1 z_2 z_3 \sim \triangle f(z_1) f(z_2) f(z_3)$
- If $z_1, z_2, z_3, z_4 \in \mathbb{C}$ are distinct points, then

$$\begin{aligned}(f(z_1), f(z_2); f(z_3), f(z_4)) &= \frac{(f(z_1) - f(z_3))(f(z_2) - f(z_4))}{(f(z_2) - f(z_3))(f(z_1) - f(z_4))} \\ &= \frac{(z_1 - z_3)(z_2 - z_4)}{(z_2 - z_3)(z_1 - z_4)} = (z_1, z_2; z_3, z_4).\end{aligned}$$

What about inversion $f(z) = \dfrac{1}{z}$? Yes, inversions also preserves cross ratios.

Complex Transformation (Optional)

Exercise 6.14

(a) Show that the signed area of any triangle is preserved under a translation or a rotation.

(b) Let $f(z) = rz$, where $r \in \mathbb{R}\backslash\{0\}$. Show that

$$\text{signed area of } \triangle f(z_1)f(z_2)f(z_3) = r^2 \cdot \text{ signed area of } \triangle z_1 z_2 z_3.$$

(c) Show that signed area of $\triangle \overline{z_1}\,\overline{z_2}\,\overline{z_3} = -$ signed area of $\triangle z_1 z_2 z_3$.

(d) Let $\varphi : \mathbb{C} \to \mathbb{C}$ be the reflection along a straight line ℓ. Show that

(i) signed area of $\triangle \varphi(z_1)\varphi(z_2)\varphi(z_3) = -$ signed area of $\triangle z_1 z_2 z_3$.

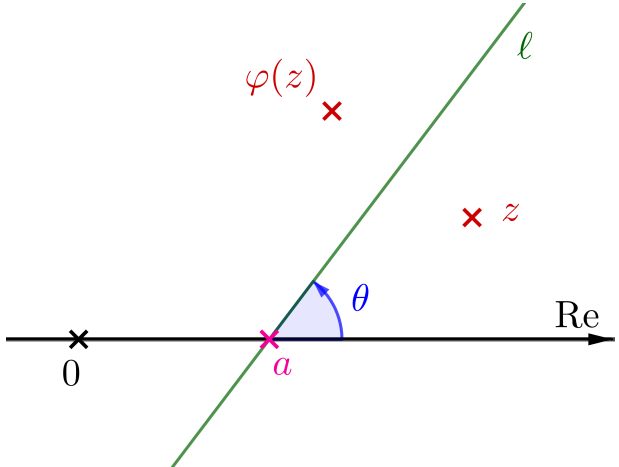
(ii) $\varphi(z_1)\varphi(z_2)\varphi(z_3) \overset{\text{rev}}{\sim} \triangle z_1 z_2 z_3$.

Hint: $\varphi = h \circ g \circ \rho \circ g^{-1} \circ h^{-1}$, where

$$\rho(z) = \bar{z},$$

$$g(z) = e^{i\theta} z \qquad g^{-1}(z) = e^{-i\theta} z$$

$$h(z) = z + a \qquad h^{-1}(z) = z - a.$$



Complex Transformation (Optional)

It is usually better to work on the extended complex plane $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$.

Definition 6.7

Let $z_1, z_2, z_3, z_4 \in \mathbb{C}_\infty$ with at least three distinct points.
The cross ratio $(z_1, z_2; z_3, z_4)$ is defined as

$$(z_1, z_2; z_3, z_4) = \begin{cases} \dfrac{(z_1 - z_3)(z_2 - z_4)}{(z_2 - z_3)(z_1 - z_4)} & \text{if } z_2 \neq z_3 \text{ and } z_1 \neq z_4 \\ \infty & \text{otherwise.} \end{cases}$$

If one of z_1, z_2, z_3, z_4 is ∞, the cross ratio is defined by taking appropriate limit. For example,

$$(z_1, z_2; z_3, \infty) = \frac{z_1 - z_3}{z_2 - z_3}.$$

Proposition 6.13

Let $z_1, z_2, z_3, z_4 \in \mathbb{C}_\infty$ with at least three distinct points. Then z_1, z_2, z_3, z_4 lies on a straight line or a circle if and only if $(z_1, z_2; z_3, z_4) \in \mathbb{R} \cup \{\infty\}$. (Convention: ∞ lies on any straight line.)

Complex Transformation (Optional)

Definition 6.8

A function $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$ is said to be a **Möbius transformation** if

$$f(z) = \frac{az+b}{cz+d} \qquad \text{where } a, b, c, d \in \mathbb{C} \text{ with } ad - bc \neq 0.$$

Note:

(1) $ad - bc \neq 0 \implies c$ and d cannot be both zero, and f is not a constant.

(2) If $c \neq 0$, we define $f(-\frac{d}{c}) = \infty$ and $f(\infty) = \frac{a}{c}$.

(3) If $c = 0$ (then $d \neq 0$), $f(z) = \frac{a}{d}z + \frac{b}{d}$ and we define $f(\infty) = \infty$.

Remark:

- The inverse of a Möbius transformation exists and is also a Möbius transformation. In fact,

$$f^{-1}(z) = \frac{dz - b}{-cz + a}.$$

- Clearly, a scaling, a translation, a rotation or an inversion is a Möbius transformation.

Proposition 6.14

Every Möbius transformation is a composition of scalings, translations, rotations and inversions.

Proof. If $c = 0$ (then $d \neq 0$), $f(z) = \frac{a}{d}z + \frac{b}{d}$, which is a rotation followed by a scaling followed by a translation.
If $c \neq 0$,

$$f(z) = \frac{az+b}{cz+d} = \frac{bc-ad}{c^2} \cdot \frac{1}{z+\frac{d}{c}} + \frac{a}{c}.$$

Let

$$\begin{aligned}
f_1(z) &= z + \frac{d}{c} && \text{(translation)} \\
f_2(z) &= \frac{1}{z} && \text{(inversion)} \\
f_3(z) &= \frac{bc-ad}{c^2} z && \text{(rotation followed by scaling)} \\
f_4(z) &= z + \frac{a}{c} && \text{(translation).}
\end{aligned}$$

Then $f(z) = (f_4 \circ f_3 \circ f_2 \circ f_1)(z)$.

Complex Transformation (Optional)

Direct consequence:

- Cross ratios are preserved by scalings, translations, rotations and inversions.
  $\implies$ Cross ratios are preserved by Möbius transformations.
- Furthermore, Möbius transformations map straight lines and circles to straight lines and circles.

Complex Transformation (Optional)

Another way to see that Möbius transformations map straight lines and circles to straight lines and circles is to consider their equations.

Lemma 6.15

The general equation of a circle or straight line in the complex plane can be written in the form

$$Az\bar{z} + \overline{B}z + B\bar{z} + C = 0 \qquad (*)$$

where $A, C \in \mathbb{R}$ *and* $|B|^2 - AC > 0$. *If* $A \neq 0$, *then* $(*)$ *represents a circle. If* $A = 0$, *then* $(*)$ *represents a straight line.*

It is then easy to check that scalings, translations, rotations and inversions preserve equation of the form $(*)$. For example, if $w = 1/z$, then $(*)$ becomes

$$\begin{aligned} & A\frac{1}{w\overline{w}} + \overline{B}\frac{1}{w} + B\frac{1}{\overline{w}} + C = 0 \\ \iff \quad & Cw\overline{w} + Bw + \overline{B}\overline{w} + A = 0. \end{aligned}$$

Hence, Möbius transformations map straight lines and circles to straight lines and circles.

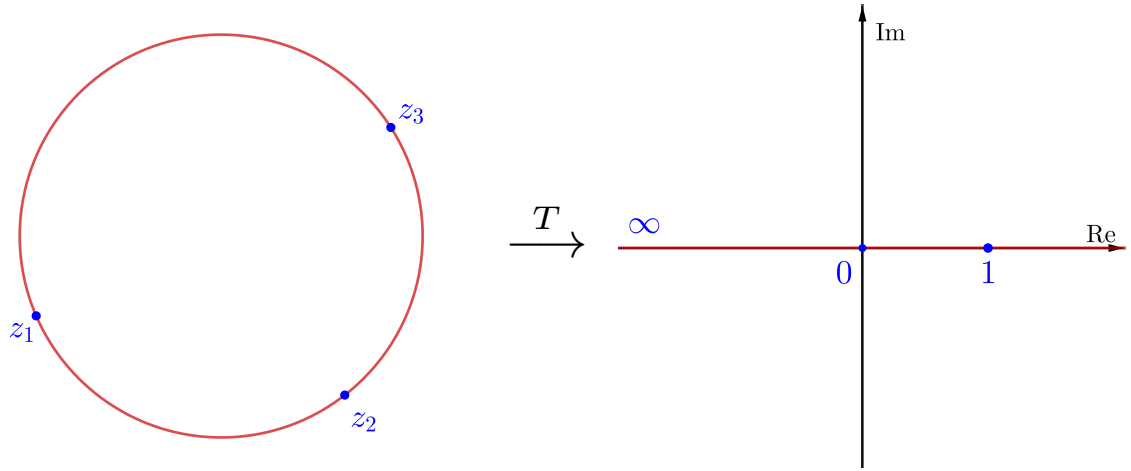
Complex Transformation (Optional)

Another look at cross ratio

Given three distinct points z_1, z_2, z_3, there is a unique Möbius transformation T which maps z_1, z_2, z_3 to $\infty, 0, 1$ respectively:

$$T(z) = \frac{(z_1 - z_3)(z_2 - z)}{(z_2 - z_3)(z_1 - z)}.$$

(Note T satisfies "$ad - bc \neq 0$" because $(z_1 - z_2), (z_2 - z_3), (z_3 - z_1) \neq 0$.)



Now the cross ratio $(z_1, z_2; z_3, z_4)$ is the image of z_4 under T:

$$(z_1, z_2; z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_2 - z_3)(z_1 - z_4)} = T(z_4).$$